

COMPUTER NETWORKS PROJECT

MILESTONE 2: CLIENT IMPLEMENTATION REVIEW

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Module Code: CMPG325

1. Introduction

This document presents the implementation and testing of the proposed enterprise network developed in Cisco Packet Tracer for Milestone 2. The purpose of this milestone was to implement the required network infrastructure, configure the assigned networking feature, verify connectivity between the different network segments, and provide evidence that the implemented network functions correctly.

The network was designed using a hierarchical architecture consisting of a Core switch, Distribution switches, Access switches, an edge router, servers, client computers and wireless access points. The network was segmented into separate VLANs to logically separate Staff, Guest, Server and Management traffic.

The implementation used the address space 10.55.0.0/16, which was sub netted into separate /24 networks for each VLAN.

The following VLANs were implemented:

VLAN	NAME	NETWORK	DEFAULT GATEWAY
10	STAFF	10.55.1.0/24	10.55.1.1
20	GUEST	10.55.2.0/24	10.55.2.1
30	SERVER	10.55.3.0/24	10.55.3.1
40	MANAGEMENT	10.55.4.0/24	10.55.4.1

The implementation also included 802.1Q trunking, inter-VLAN routing, DHCP, Port Security, Guest VLAN isolation and router connectivity.

2. Network Topology

The physical network uses an Extended Star Topology.

The Cisco 3650 Core switch acts as the central connection point. It connects to four Distribution switches, with each Distribution switch representing one floor of the network. Each Distribution switch connects to its corresponding Access switch, which then connects to the end devices on that floor.

The network structure is:

ISR4331 Router → Core-3650 → Distribution Switches → Access Switches → End Devices

The Extended Star Topology was selected because it provides a structured network layout that can be expanded by adding additional switches or devices without redesigning the entire network.

VLANs were used to provide logical segmentation within this physical topology. The physical topology therefore provides the connectivity structure, while the VLANs separate network traffic according to the function of the devices.

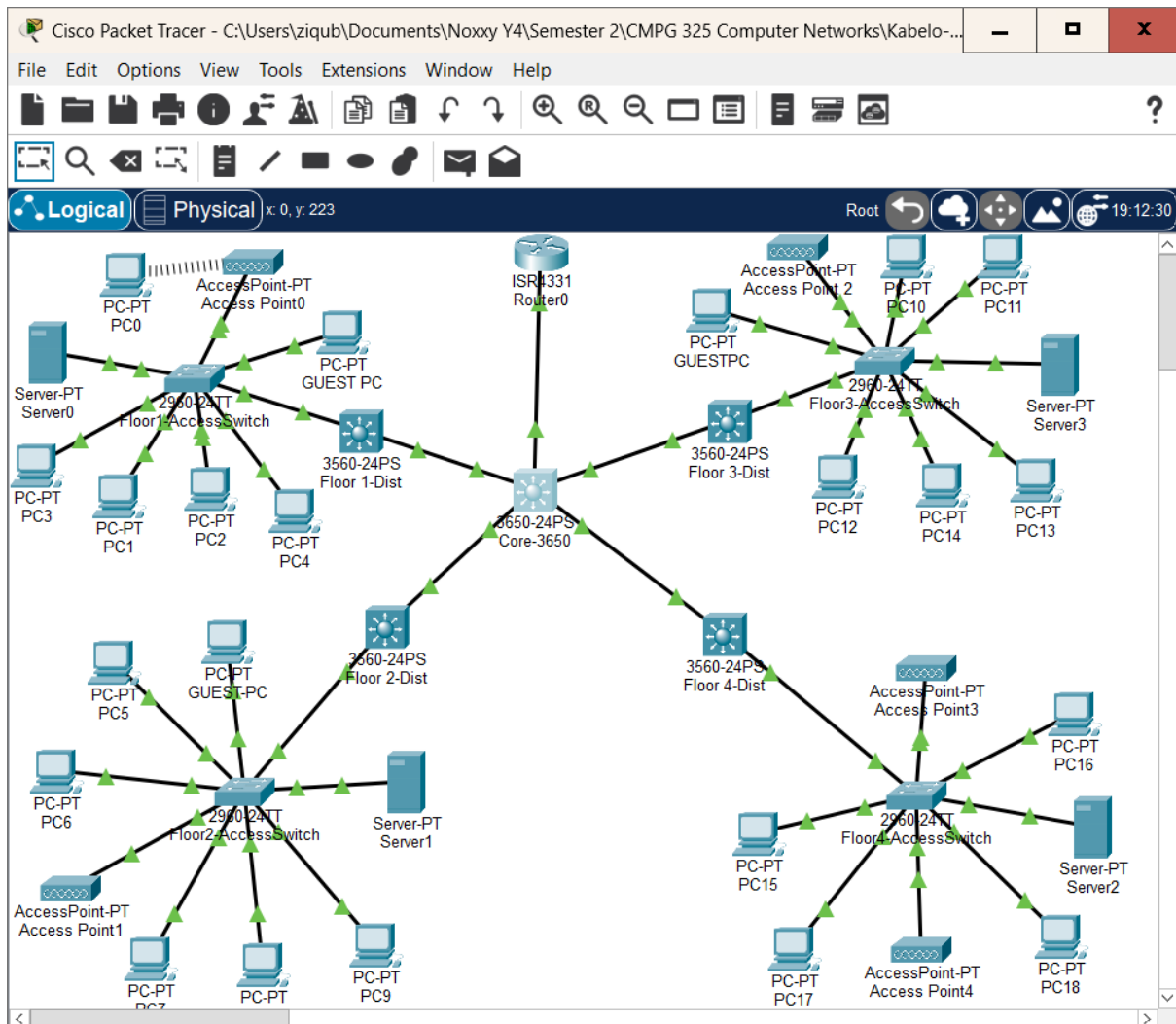


FIGURE:1 Final Enterprise Network Extended Star Topology in Cisco Packet Tracer.

3. VLAN Implementation

Four VLANs were implemented throughout the switching infrastructure.

VLAN 10 was assigned to Staff devices, VLAN 20 was assigned to Guest devices, VLAN 30 was assigned to Servers and VLAN 40 was assigned to Management devices.

This segmentation separates different types of network users and resources into different broadcast domains.

Staff devices on Floors 1 to 3 were placed in VLAN 10, while Management devices on Floor 4 were placed in VLAN 40. Servers across all four floors were placed in VLAN 30. Guest PCs on Floors 1, 2 and 3 were placed in VLAN 20.

The VLAN configuration was applied across the required switches so that the same VLANs could be carried throughout the network.

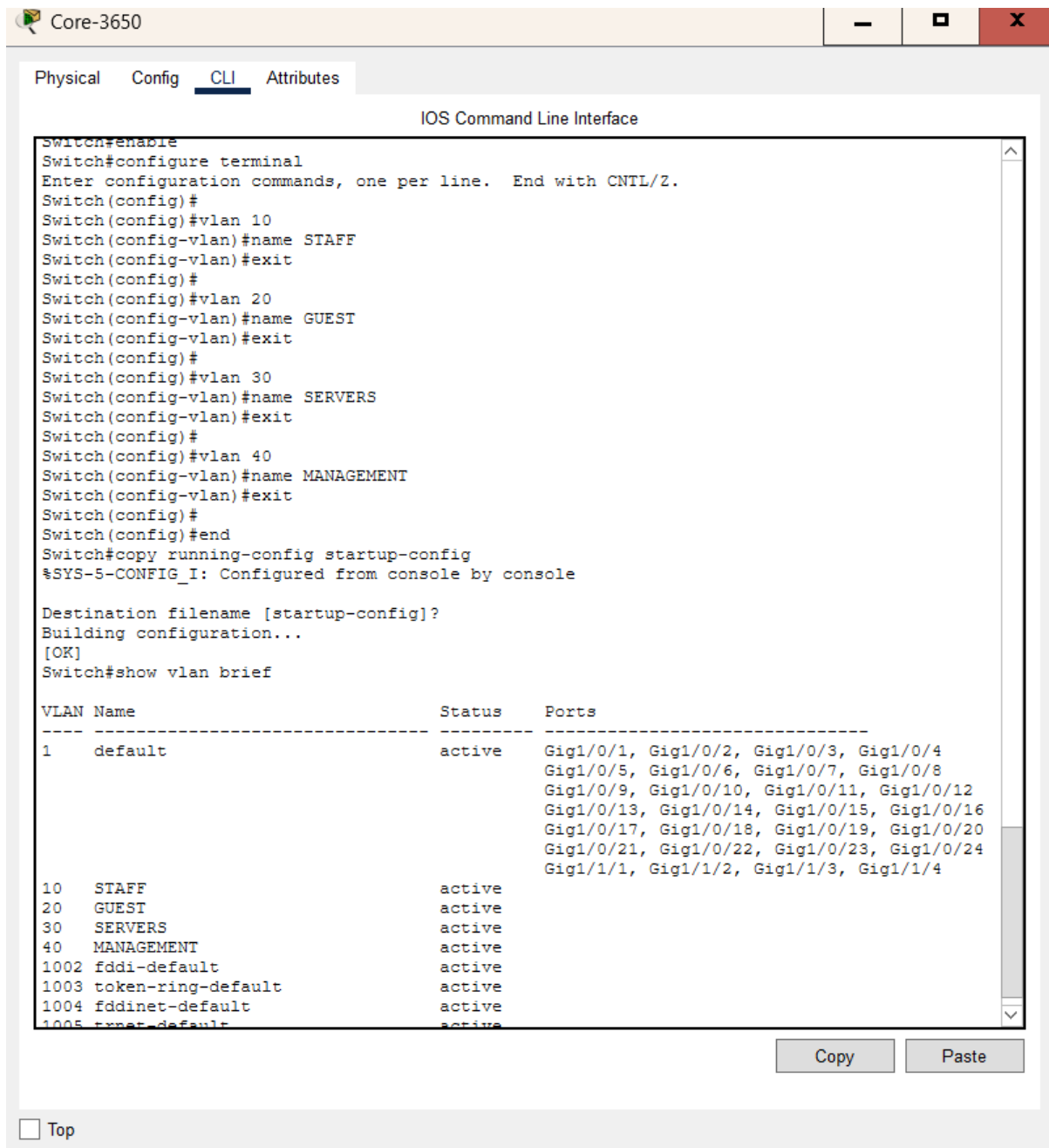


FIGURE2: Core-3650 VLAN Configuration and Verification

Floor 1-Dist

Physical

Config

CLI

Attributes

IOS Command Line Interface

```

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#vlan 10
Switch(config-vlan)#name STAFF
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 20
Switch(config-vlan)#name GUEST
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 30
Switch(config-vlan)#name SERVERS
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#vlan 40
Switch(config-vlan)#name MANAGEMENT
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#end
Switch#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	STAFF	active	
20	GUEST	active	
30	SERVERS	active	
40	MANAGEMENT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

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FIGURE3: Floor 1 Distribution Switch VLAN Configuration

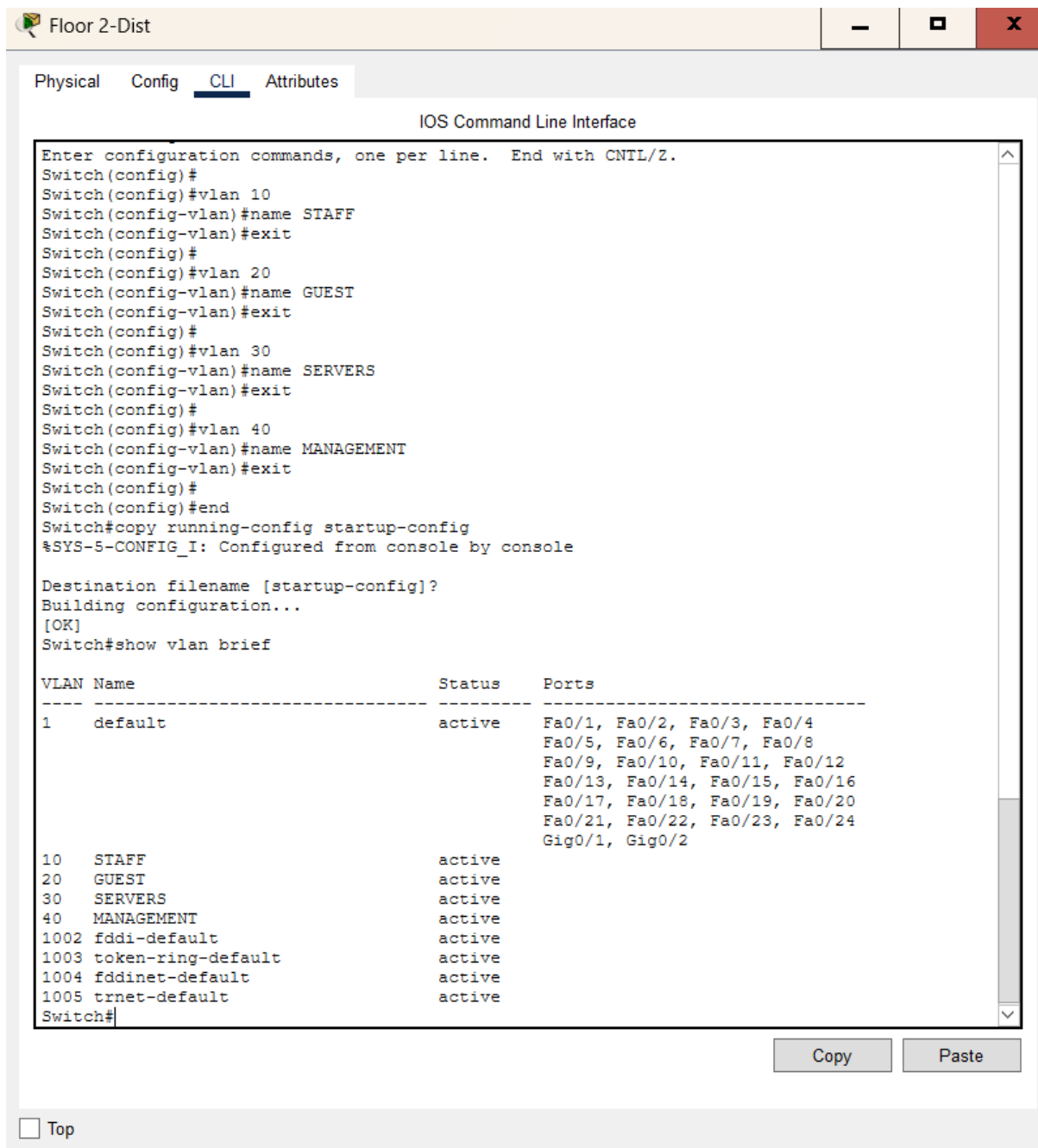


FIGURE 4: Floor 2 Distribution Switch VLAN Configuration

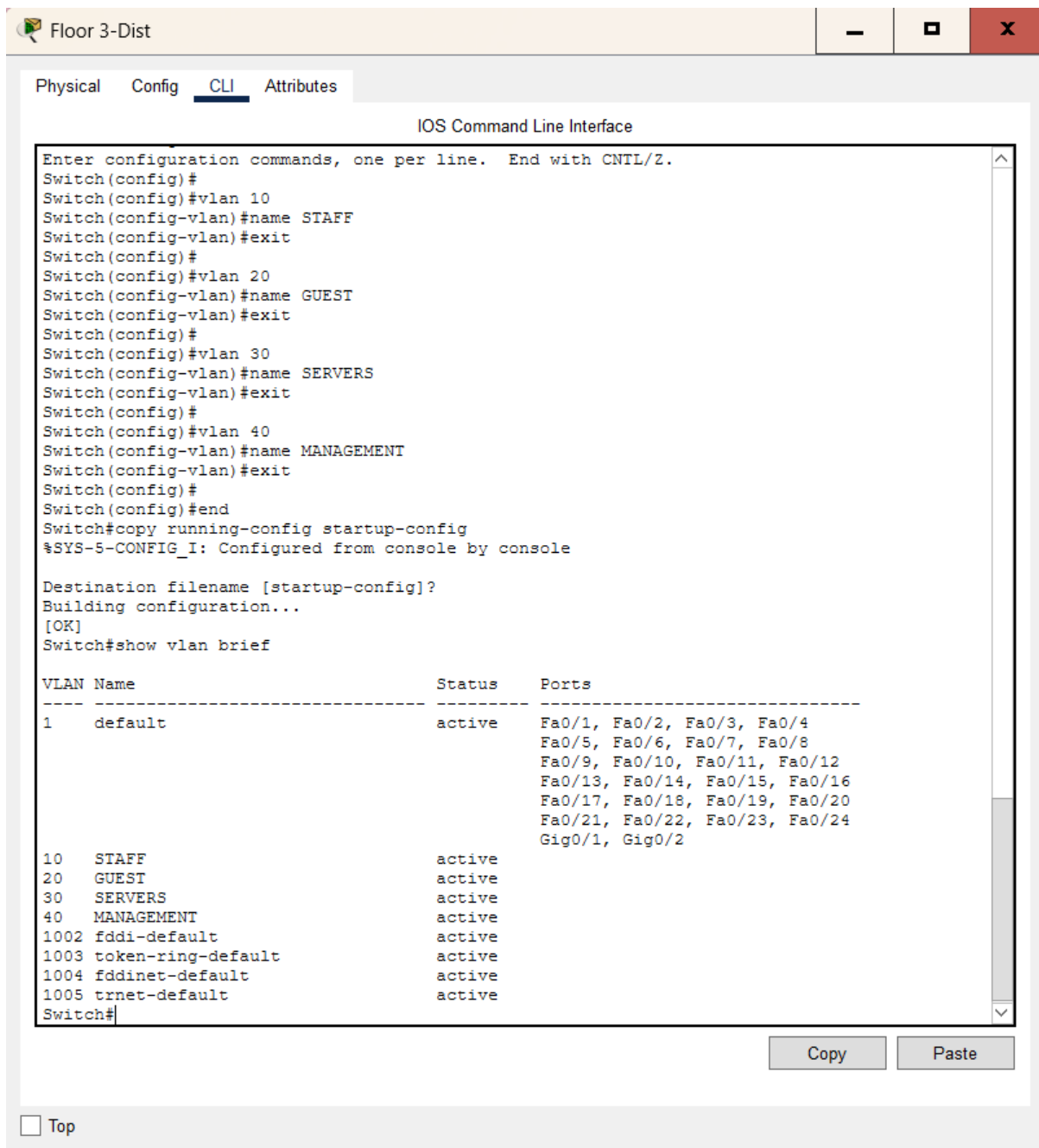


FIGURE 5: Floor 3 Distribution Switch VLAN Configuration

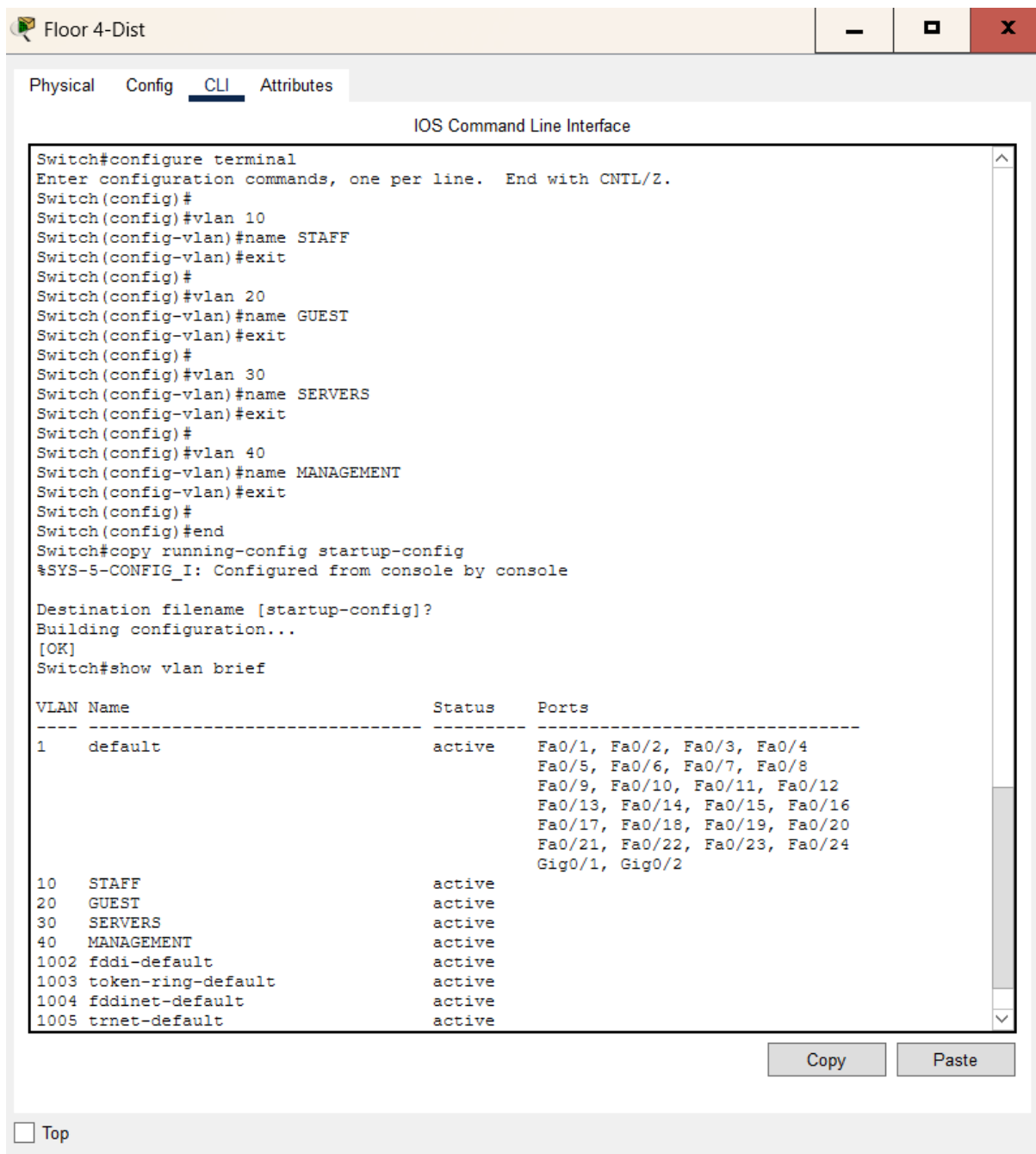


FIGURE 6: Floor 4 Distribution Switch VLAN Configuration

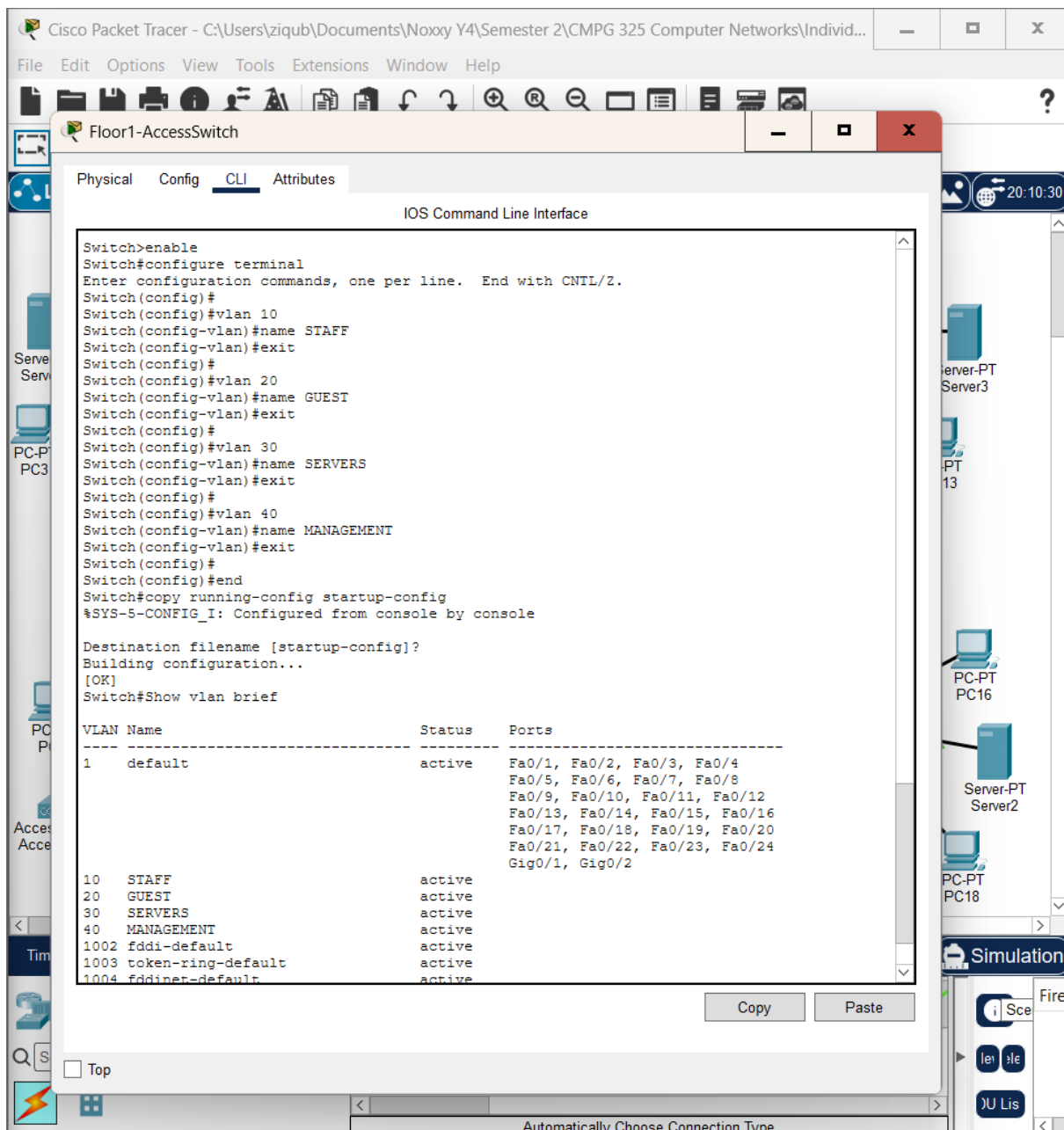


FIGURE 7: Floor Access Switch VLAN Configuration (Same code was run on Floor 2, 3 and 4.)

- Access Port VLANs

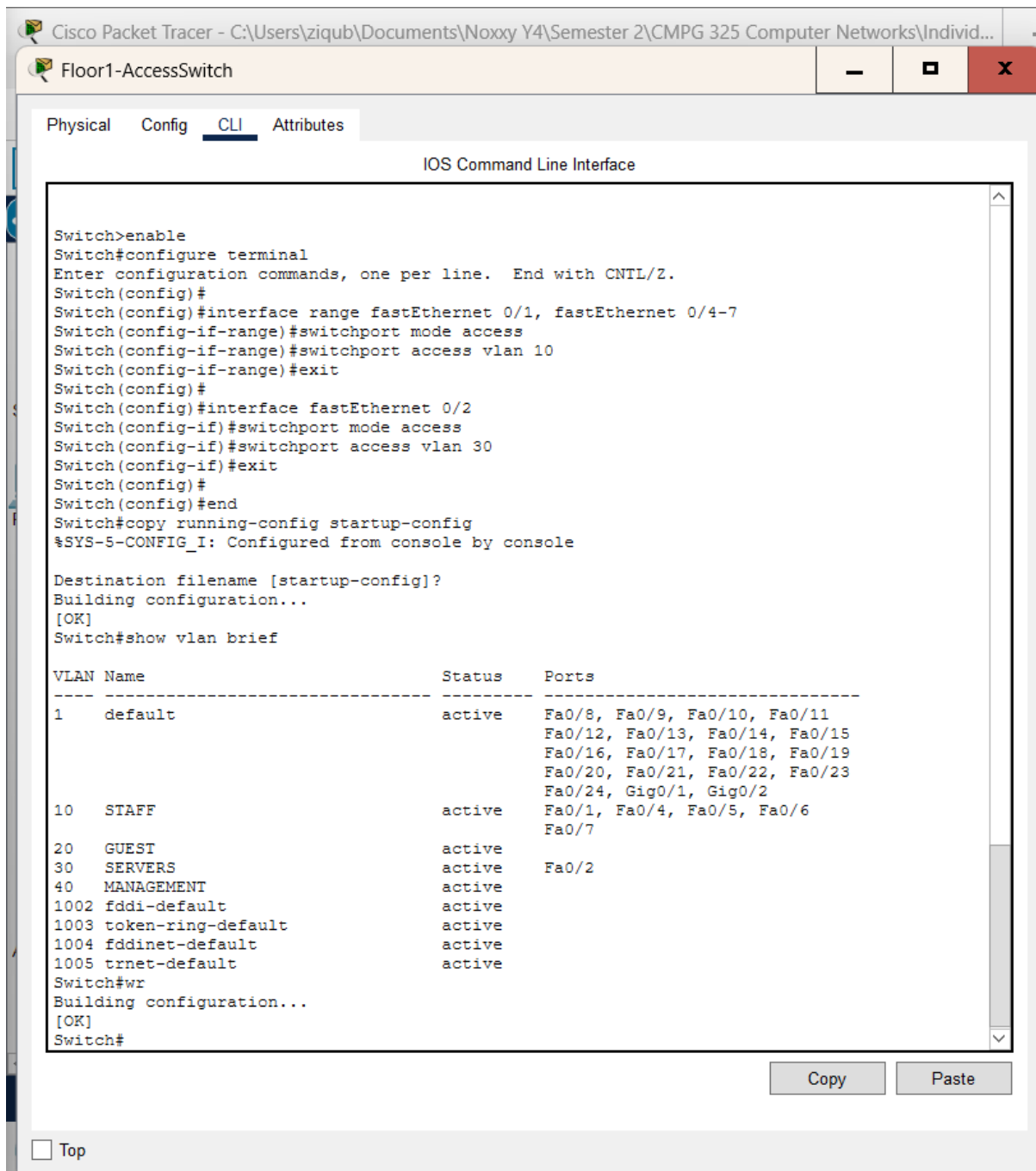


FIGURE 8: Floor 1 Access Port VLAN Assignment (Staff and Server VLANs to user-facing access ports on the Floor 1 Access Switch.)

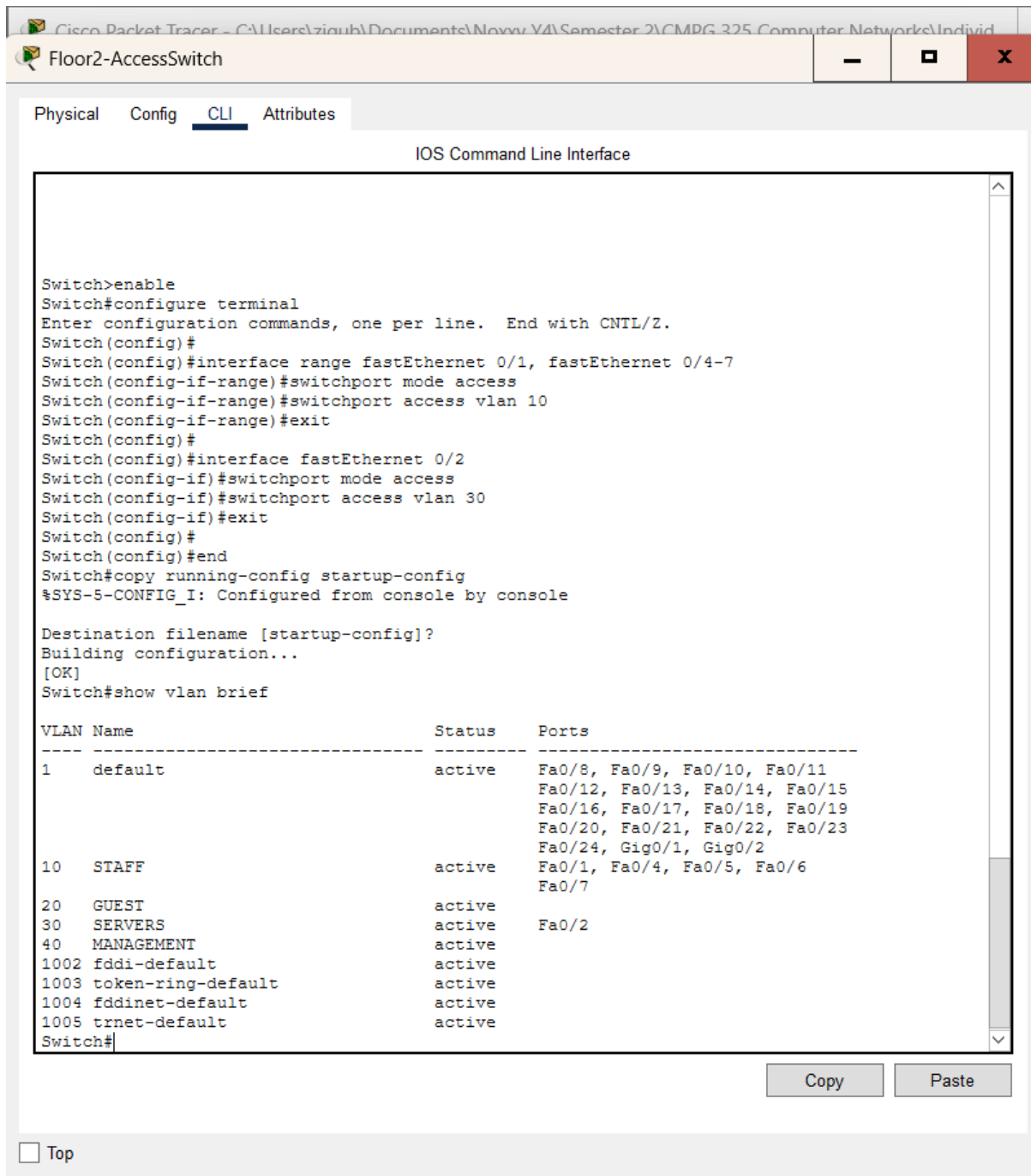


FIGURE 9: Floor 2 Access Port VLAN Assignment

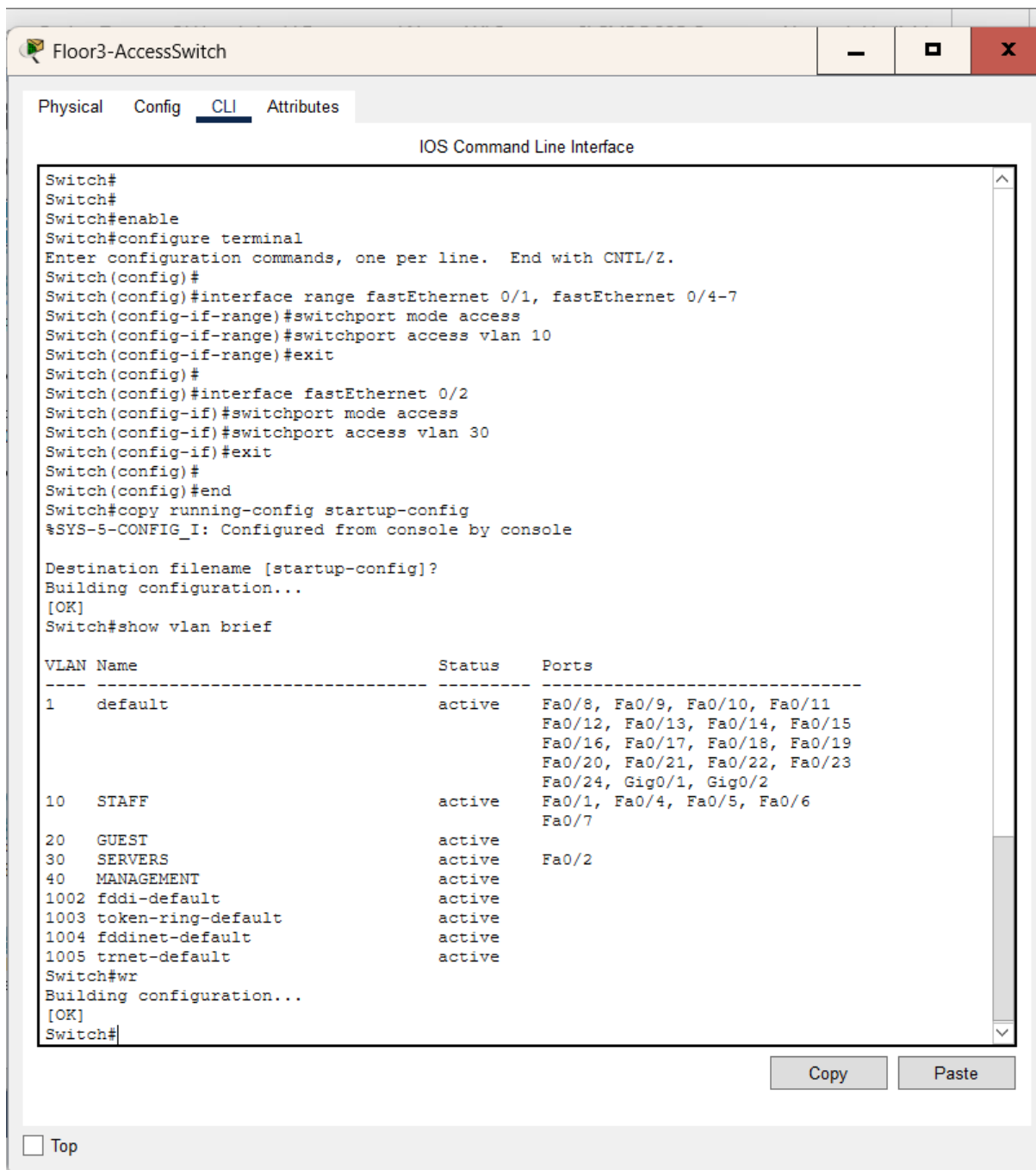


FIGURE 10: Floor 3 Access Port VLAN Assignment

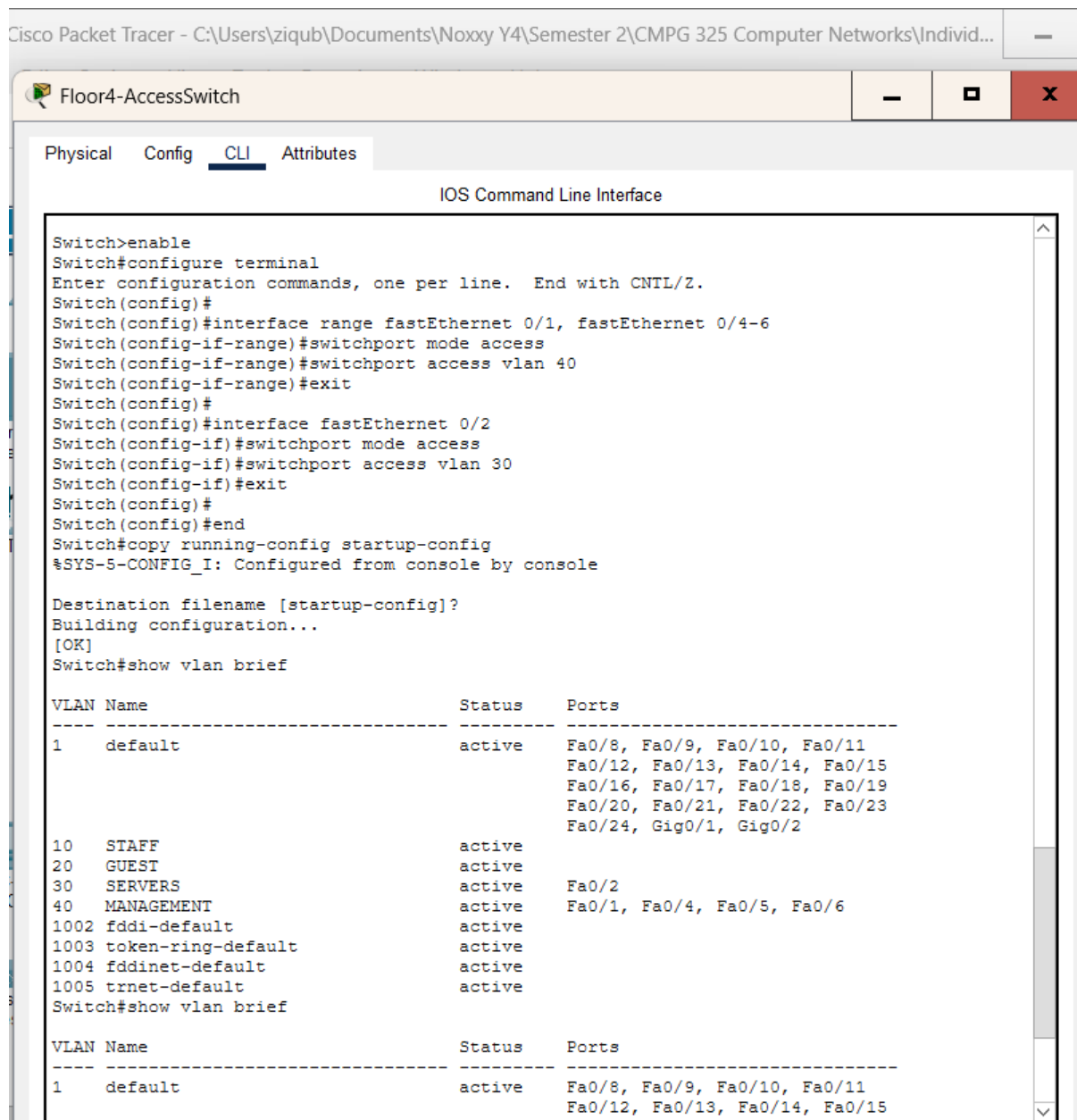


FIGURE 11: Floor 4 Access Port VLAN Assignment

4. Trunk Link Implementation

Trunk links were configured to allow traffic from multiple VLANs to travel between the different layers of the network.

Trunking was implemented between the Core and Distribution switches and between the Distribution and Access switches.

The trunk links were configured to carry the required VLANs:

- VLAN 10 — Staff
- VLAN 20 — Guest
- VLAN 30 — Servers
- VLAN 40 — Management

The Access switch uplinks were also configured as trunks. This was important because devices connected to the Access switches needed to communicate with the appropriate VLAN gateways located on the Core switch. The trunk configuration ensures that VLAN traffic can travel through the Extended Star topology without losing its VLAN identity.

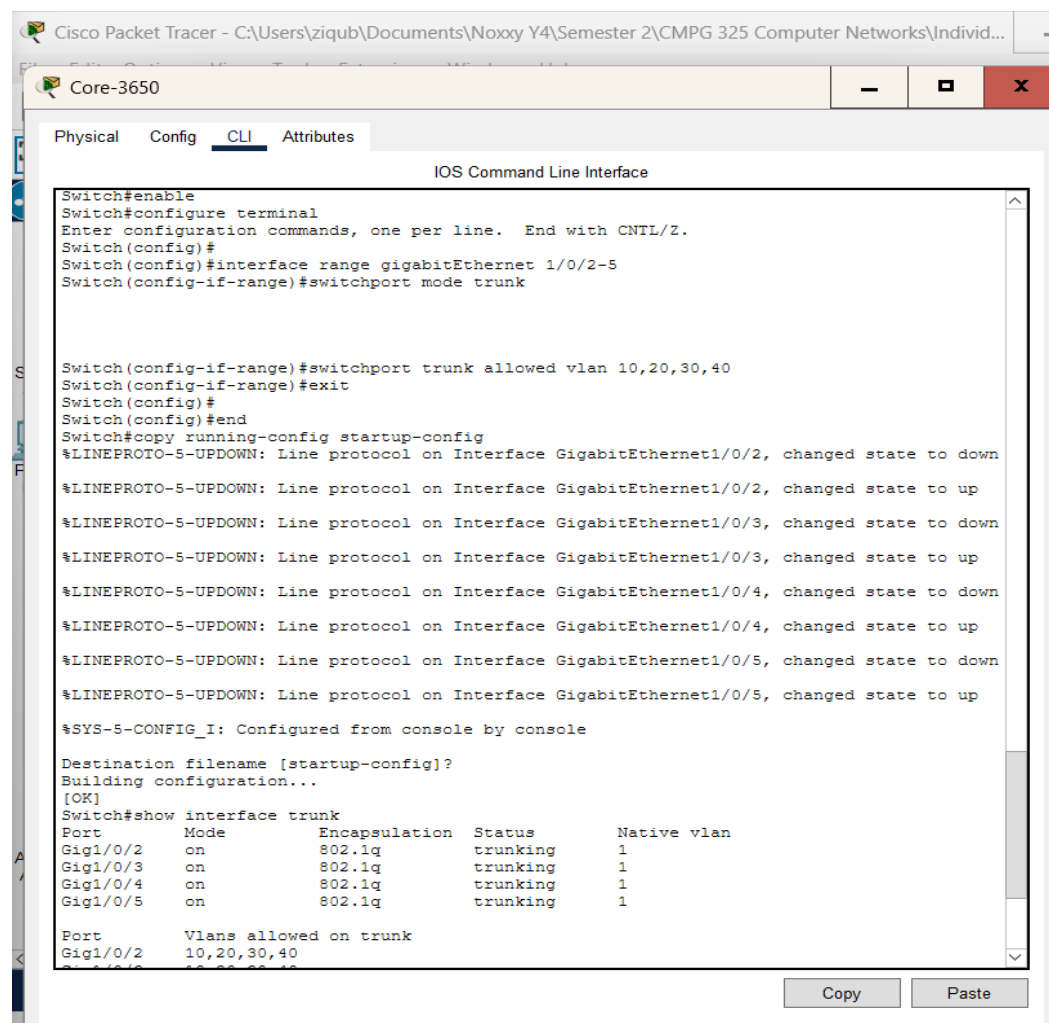


FIGURE 12: Core-3650 Trunk Port Configuration

Floor 1-Dist

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to
"trunk" mode.
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    auto      n-802.1q       trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,30,40

Switch#
```

FIGURE 13: Floor 1 Distribution Switch Trunk Verification

Floor 2-Dist

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to
"trunk" mode.
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    auto      n-802.1q       trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,30,40

Switch#wr
Building configuration...
[OK]
```

FIGURE 14: Floor 2 Distribution Switch Trunk Verification

Floor 3-Dist

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to
"trunk" mode.
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    auto      n-802.1q       trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,30,40

Switch#wr
Building configuration...
[OK]
```

FIGURE 15: Floor 3 Distribution Switch Trunk Verification

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Floor 4-Dist

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface gigabitEthernet 0/2
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to
"trunk" mode.
Switch(config-if)#switchport trunk allowed vlan 10,20,30,40
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    auto      n-802.1q       trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    1,10,20,30,40

Switch#wr
Building configuration...
[OK]
Switch#
```

FIGURE 16: Floor 4 Distribution Switch Trunk Verification

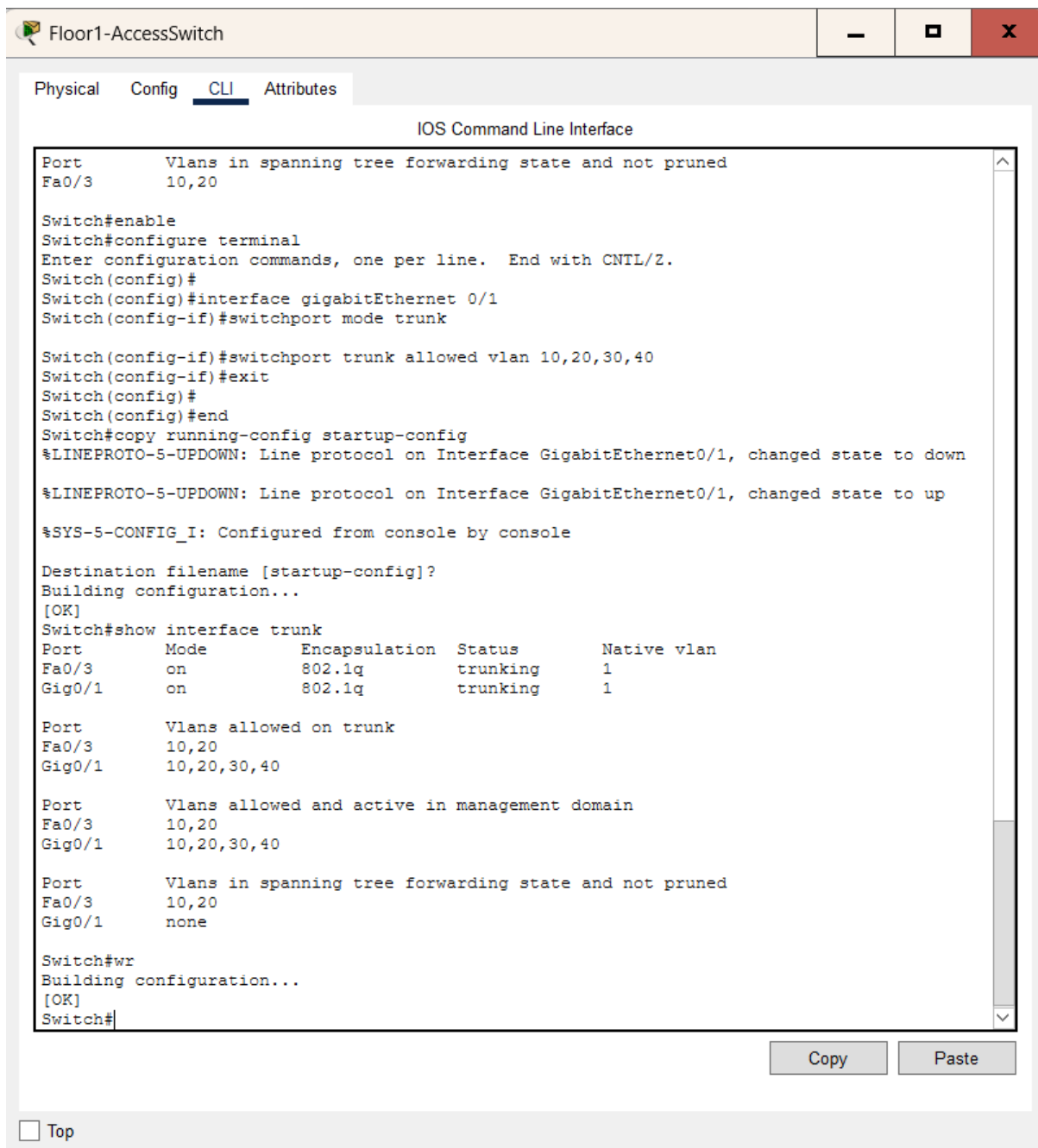


FIGURE17: Floor 1 Access Switch Uplink Trunk Configuration

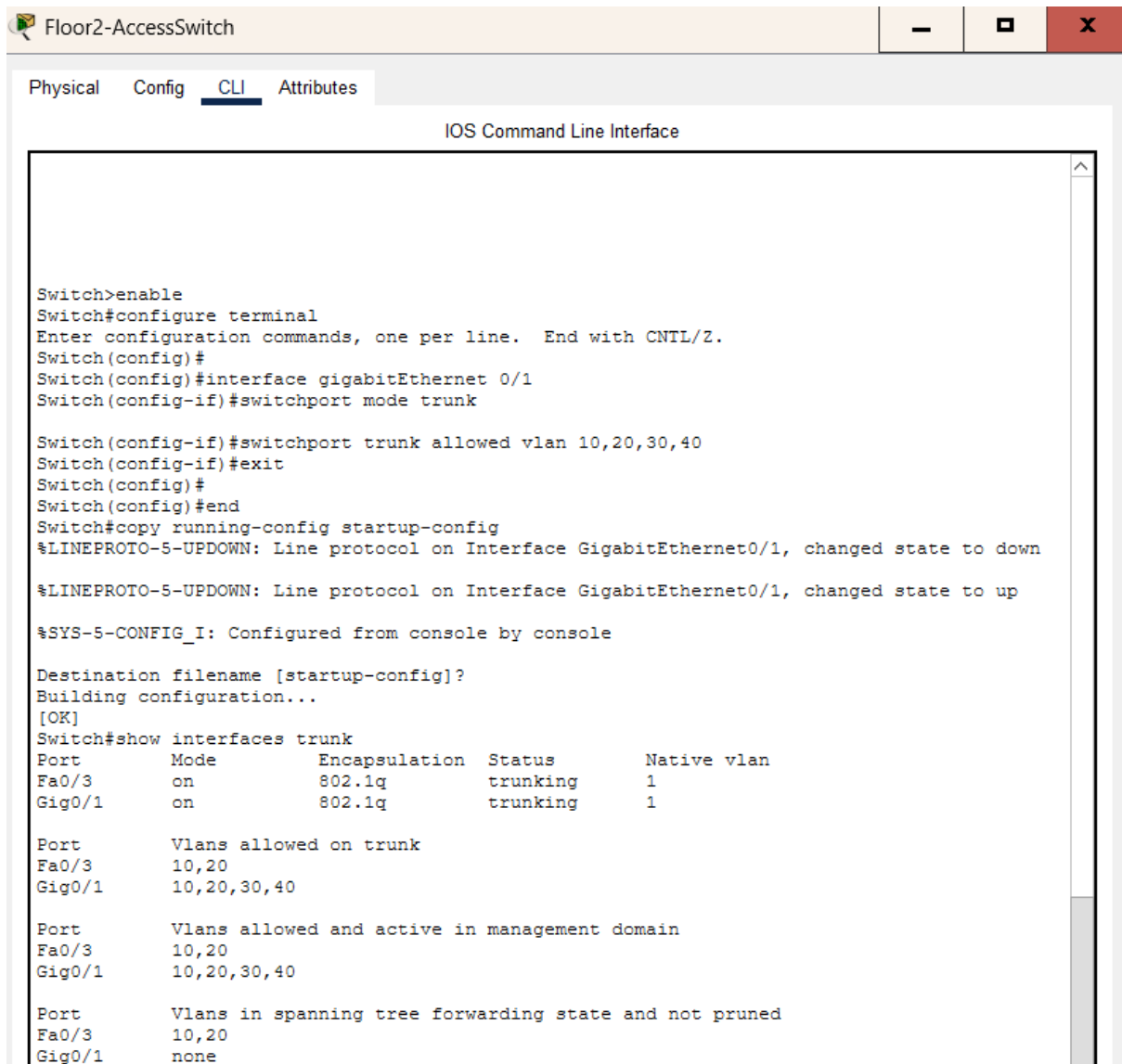


FIGURE 18: Floor 2 Access Switch Uplink Trunk Configuration

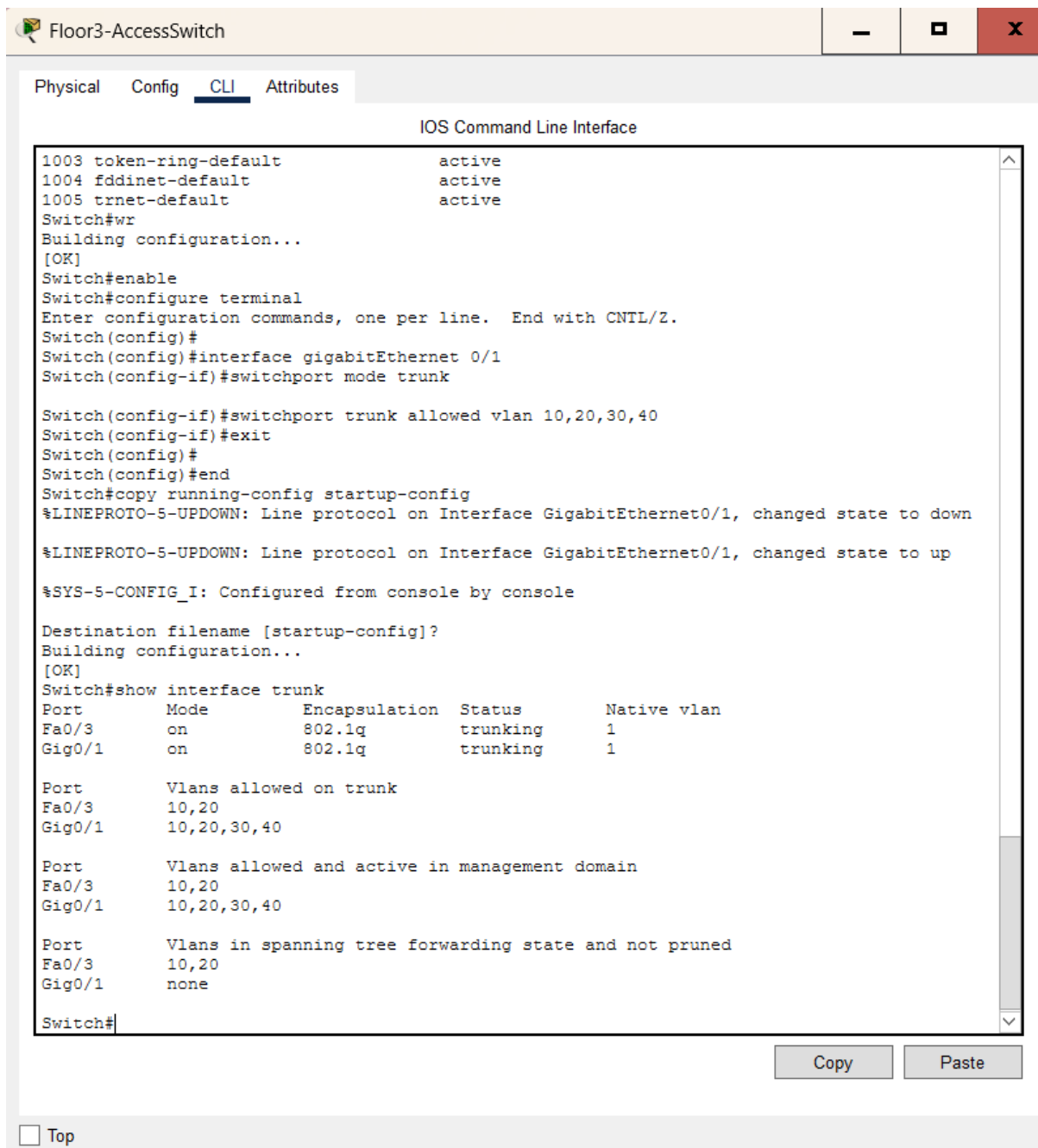


FIGURE 19: Floor 3 Access Switch Uplink Trunk Configuration

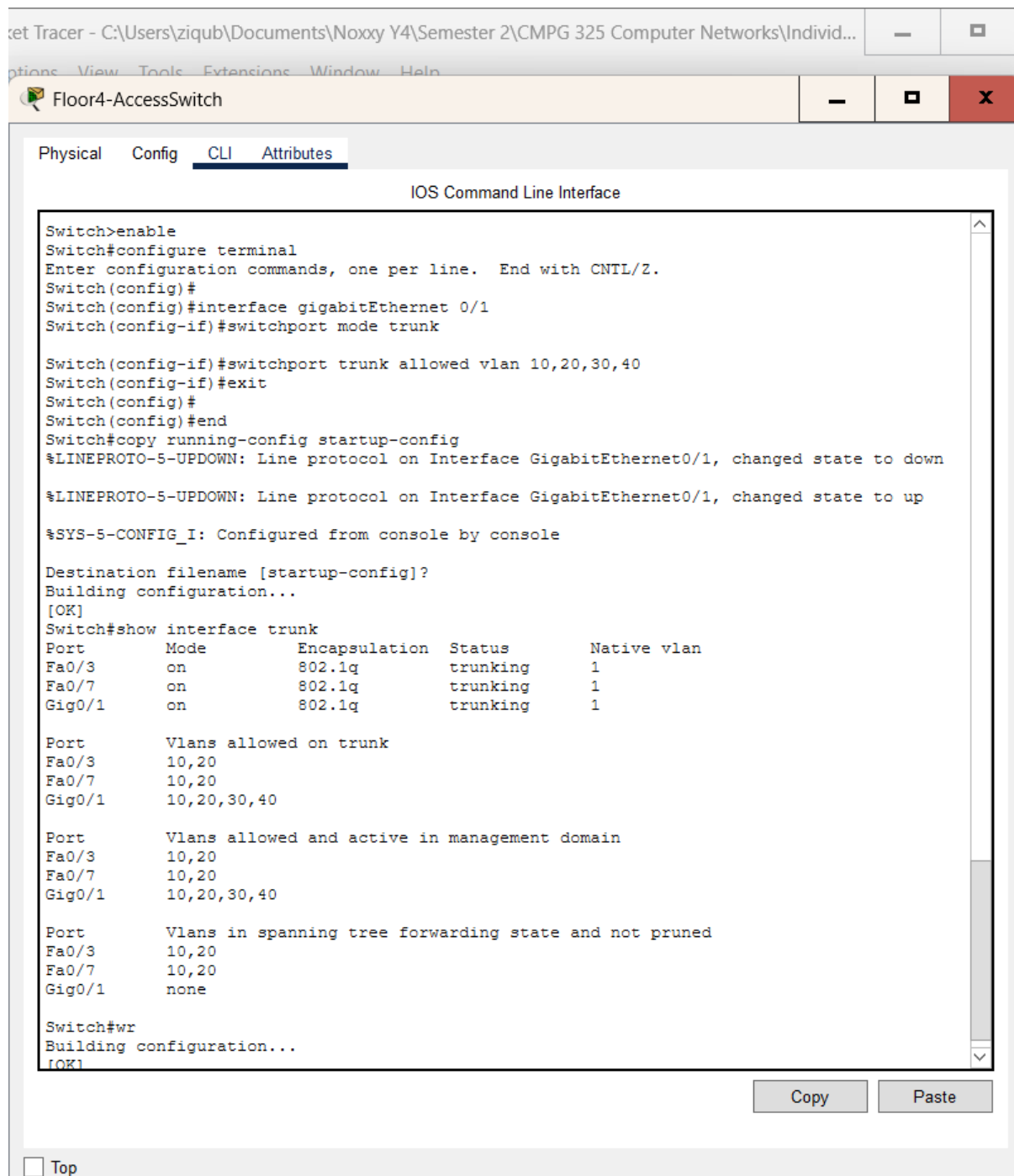


FIGURE 20: Floor 4 Access Switch Uplink Trunk Configuration

5. Inter-VLAN Routing Implementation

Inter-VLAN routing was implemented on the Core-3650 switch.

Each VLAN was assigned a Switch Virtual Interface (SVI), which acts as the default gateway for devices in that VLAN.

The configured gateways were:

- Staff — 10.55.1.1
- Guest — 10.55.2.1
- Servers — 10.55.3.1
- Management — 10.55.4.1

IP routing was enabled on the Core switch to allow communication between different VLANs.

This configuration allows devices belonging to different VLANs to communicate when the traffic is permitted. For example, a Staff PC can communicate with a Management PC or a Server through the Core switch.

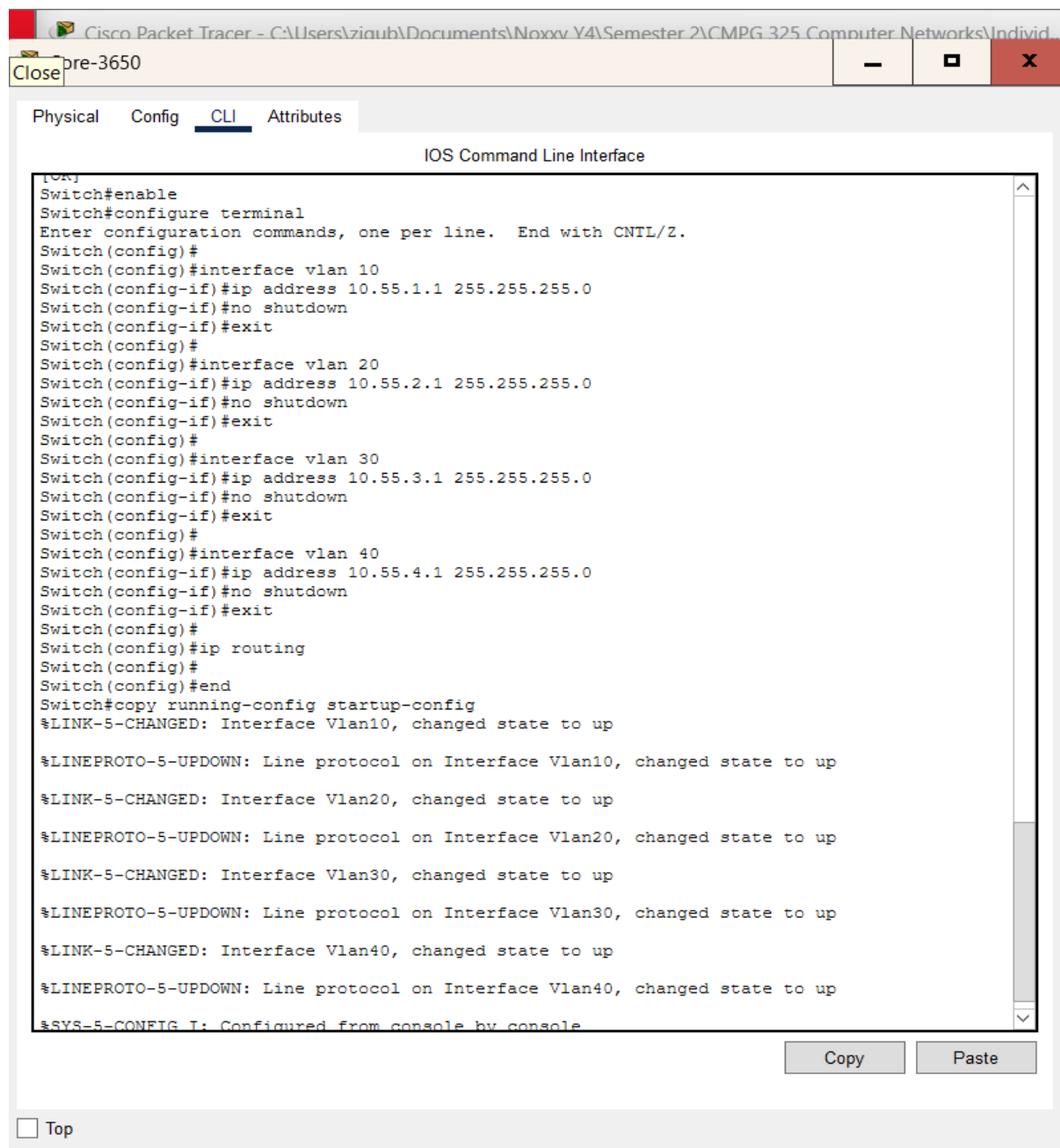


FIGURE 21: VLAN Gateway (SVI) Configuration on Core-3650

```

Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet1/0/1	unassigned	YES	unset	up	up
GigabitEthernet1/0/2	unassigned	YES	unset	up	up
GigabitEthernet1/0/3	unassigned	YES	unset	up	up
GigabitEthernet1/0/4	unassigned	YES	unset	up	up
GigabitEthernet1/0/5	unassigned	YES	unset	up	up
GigabitEthernet1/0/6	unassigned	YES	unset	down	down
GigabitEthernet1/0/7	unassigned	YES	unset	down	down
GigabitEthernet1/0/8	unassigned	YES	unset	down	down
GigabitEthernet1/0/9	unassigned	YES	unset	down	down
GigabitEthernet1/0/10	unassigned	YES	unset	down	down
GigabitEthernet1/0/11	unassigned	YES	unset	down	down
GigabitEthernet1/0/12	unassigned	YES	unset	down	down
GigabitEthernet1/0/13	unassigned	YES	unset	down	down
GigabitEthernet1/0/14	unassigned	YES	unset	down	down
GigabitEthernet1/0/15	unassigned	YES	unset	down	down
GigabitEthernet1/0/16	unassigned	YES	unset	down	down
GigabitEthernet1/0/17	unassigned	YES	unset	down	down
GigabitEthernet1/0/18	unassigned	YES	unset	down	down
GigabitEthernet1/0/19	unassigned	YES	unset	down	down
GigabitEthernet1/0/20	unassigned	YES	unset	down	down
GigabitEthernet1/0/21	unassigned	YES	unset	down	down
GigabitEthernet1/0/22	unassigned	YES	unset	down	down
GigabitEthernet1/0/23	unassigned	YES	unset	down	down
GigabitEthernet1/0/24	unassigned	YES	unset	down	down
GigabitEthernet1/1/1	unassigned	YES	unset	down	down
GigabitEthernet1/1/2	unassigned	YES	unset	down	down
GigabitEthernet1/1/3	unassigned	YES	unset	down	down
GigabitEthernet1/1/4	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	10.55.1.1	YES	manual	up	up
Vlan20	10.55.2.1	YES	manual	up	up
Vlan30	10.55.3.1	YES	manual	up	up
Vlan40	10.55.4.1	YES	manual	up	up

```

Switch#

```

FIGURE 22: SVI Interface Status Verification

```

Switch#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.55.1.0/24 is directly connected, Vlan10
L       10.55.1.1/32 is directly connected, Vlan10
C       10.55.2.0/24 is directly connected, Vlan20
L       10.55.2.1/32 is directly connected, Vlan20
C       10.55.3.0/24 is directly connected, Vlan30
L       10.55.3.1/32 is directly connected, Vlan30
C       10.55.4.0/24 is directly connected, Vlan40
L       10.55.4.1/32 is directly connected, Vlan40

Switch#

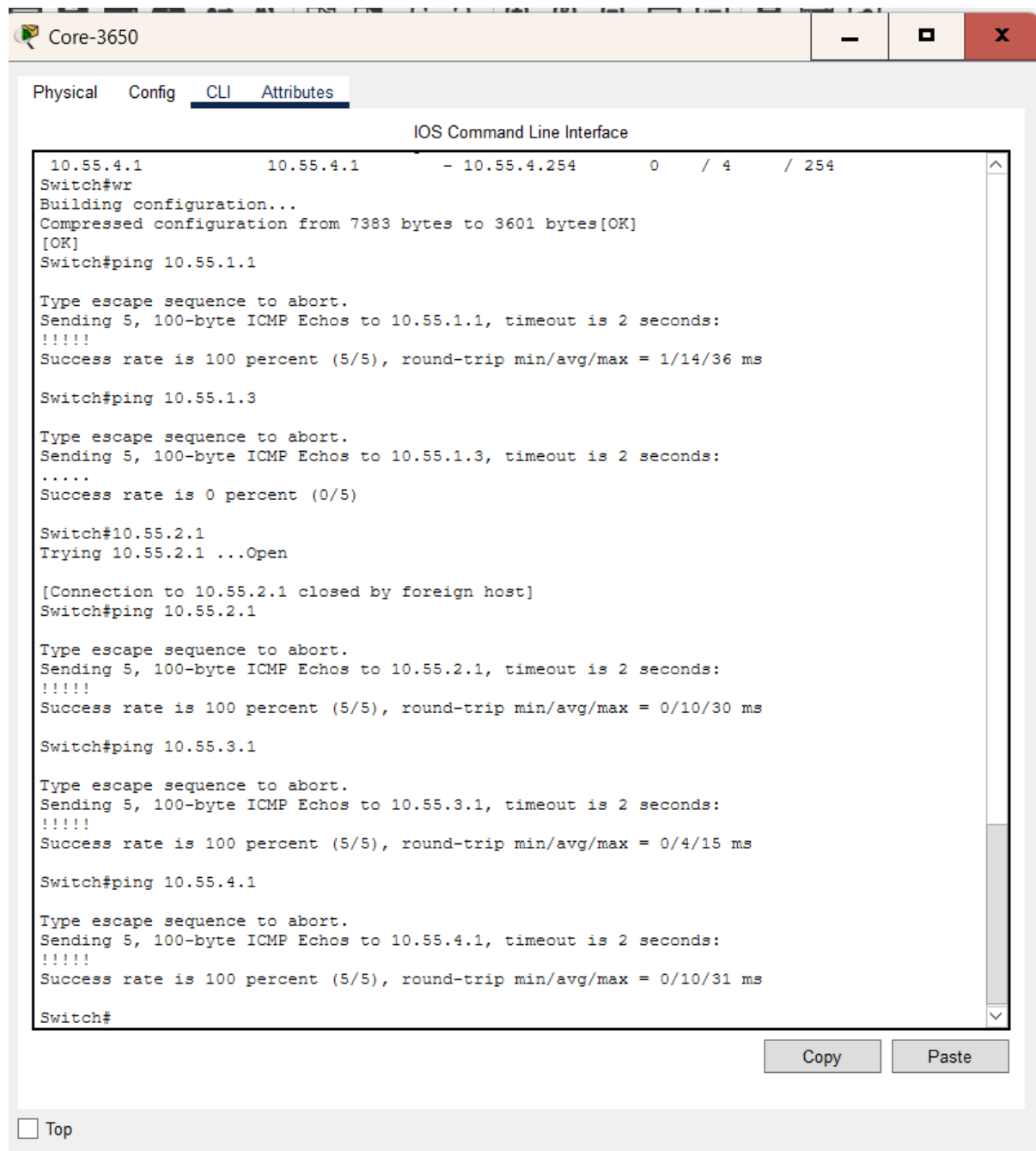
```

FIGURE 23: Core-3650 Routing Table Verification

- **Inter-VLAN Connectivity Test**

Connectivity was tested between devices belonging to different VLANs.

A Staff PC was used to test communication with the Management network and Server network.



The screenshot shows a terminal window titled "Core-3650" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The output shows the following commands and results:

```
10.55.4.1      10.55.4.1      - 10.55.4.254    0    / 4    / 254
Switch#wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Switch#ping 10.55.1.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/14/36 ms

Switch#ping 10.55.1.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.1.3, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Switch#10.55.2.1
Trying 10.55.2.1 ...Open

[Connection to 10.55.2.1 closed by foreign host]
Switch#ping 10.55.2.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/10/30 ms

Switch#ping 10.55.3.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/4/15 ms

Switch#ping 10.55.4.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.4.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/10/31 ms

Switch#
```

At the bottom of the window, there is a "Top" button and a "Copy/Paste" section with "Copy" and "Paste" buttons.

FIGURE 24

Inter-VLAN Routing Connectivity Test

Ping and connectivity tests confirming successful communication between all departmental VLANs.

```
C:\>ping 10.55.4.2

Pinging 10.55.4.2 with 32 bytes of data:

Request timed out.
Reply from 10.55.4.2: bytes=32 time<1ms TTL=127
Reply from 10.55.4.2: bytes=32 time=1ms TTL=127
Reply from 10.55.4.2: bytes=32 time=1ms TTL=127

Ping statistics for 10.55.4.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.55.4.2

Pinging 10.55.4.2 with 32 bytes of data:

Reply from 10.55.4.2: bytes=32 time<1ms TTL=127
Reply from 10.55.4.2: bytes=32 time<1ms TTL=127
Reply from 10.55.4.2: bytes=32 time<1ms TTL=127
Reply from 10.55.4.2: bytes=32 time<1ms TTL=127

Ping statistics for 10.55.4.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

FIGURE 25: Successful Inter-VLAN Communication Between Staff(PC) and Management VLANs(PC)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.55.3.3

Pinging 10.55.3.3 with 32 bytes of data:

Request timed out.
Reply from 10.55.3.3: bytes=32 time<1ms TTL=127
Reply from 10.55.3.3: bytes=32 time<1ms TTL=127
Reply from 10.55.3.3: bytes=32 time=1ms TTL=127

Ping statistics for 10.55.3.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.55.3.3

Pinging 10.55.3.3 with 32 bytes of data:

Reply from 10.55.3.3: bytes=32 time<1ms TTL=127
Reply from 10.55.3.3: bytes=32 time<1ms TTL=127
Reply from 10.55.3.3: bytes=32 time<1ms TTL=127
Reply from 10.55.3.3: bytes=32 time<1ms TTL=127

Ping statistics for 10.55.3.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

FIGURE 26: Successful Inter-VLAN Communication Between Staff(PC) and Server VLANs

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.55.1.9

Pinging 10.55.1.9 with 32 bytes of data:

Reply from 10.55.1.9: bytes=32 time=6ms TTL=128
Reply from 10.55.1.9: bytes=32 time<1ms TTL=128
Reply from 10.55.1.9: bytes=32 time=13ms TTL=128
Reply from 10.55.1.9: bytes=32 time=12ms TTL=128

Ping statistics for 10.55.1.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 7ms

C:\>
```

FIGURE 27: Successful Inter-VLAN Communication Between Staff(PC) and Staff(PC) VLANs

6. **DHCP Implementation**

DHCP was configured on the Core switch to provide automatic IP addressing to client devices.

Separate DHCP pools were created for:

- Staff
- Guest
- Servers
- Management

Each pool was associated with the correct network and default gateway.

Addresses used by the VLAN gateways and reserved server addresses were excluded from the DHCP allocation range.

The DHCP implementation reduced the need to manually configure IP addresses on client devices and ensured that devices received addresses from the correct subnet.

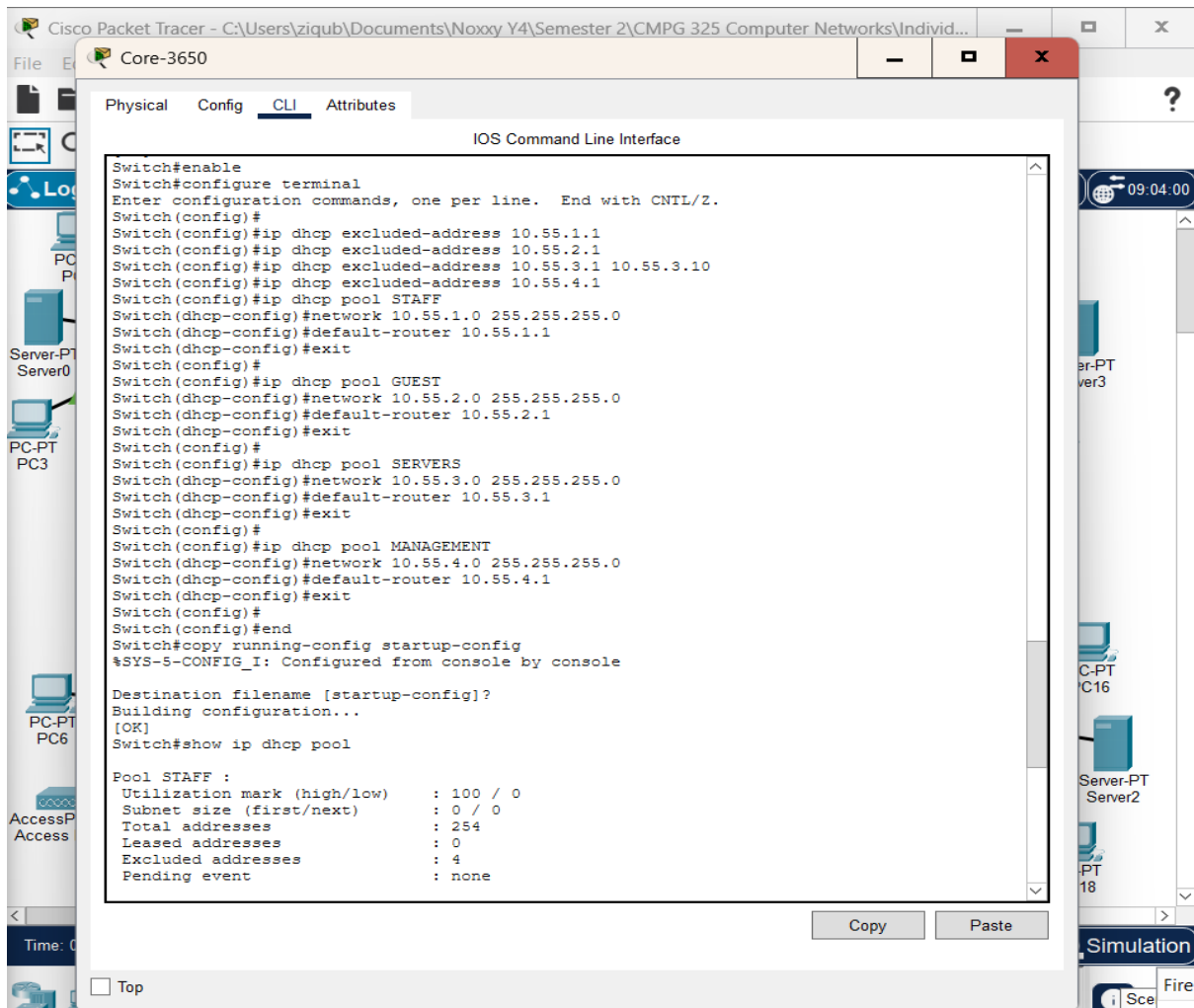


FIGURE 28: DHCP Pool Configuration on Core-3650

```

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.55.1.1          10.55.1.1 - 10.55.1.254    0 / 4 / 254

Pool GUEST :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 254
Leased addresses             : 0
Excluded addresses           : 4
Pending event                : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.55.2.1          10.55.2.1 - 10.55.2.254    0 / 4 / 254

Pool SERVERS :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 254
Leased addresses             : 0
Excluded addresses           : 4
Pending event                : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.55.3.1          10.55.3.1 - 10.55.3.254    0 / 4 / 254

Pool MANAGEMENT :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses              : 254
Leased addresses             : 0
Excluded addresses           : 4
Pending event                : none

1 subnet is currently in the pool
Current index      IP address range      Leased/Excluded/Total
10.55.4.1          10.55.4.1 - 10.55.4.254    0 / 4 / 254
Switch#

```

FIGURE 29: DHCP Pool Verification

- **DHCP Client Testing**

Client PCs were configured to obtain their addresses automatically using DHCP. The PCs on the Staff network received addresses from the 10.55.1.0/24 network, while Management PCs received addresses from the 10.55.4.0/24 network.

- DHCP address assignment was verified on all four floors. Devices successfully obtained IP addresses, subnet masks, default gateways and DNS information from the appropriate VLAN DHCP pool. Figures 30 to 35 show successful DHCP allocation on Floors 1, 2, 3 and 4.

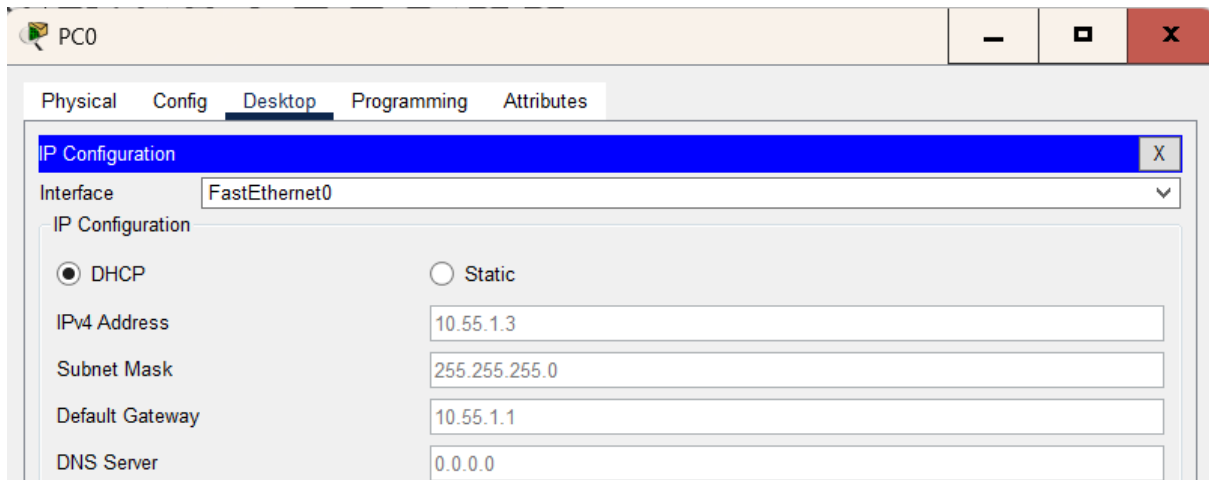


FIGURE 30: FLOOR 1 PC

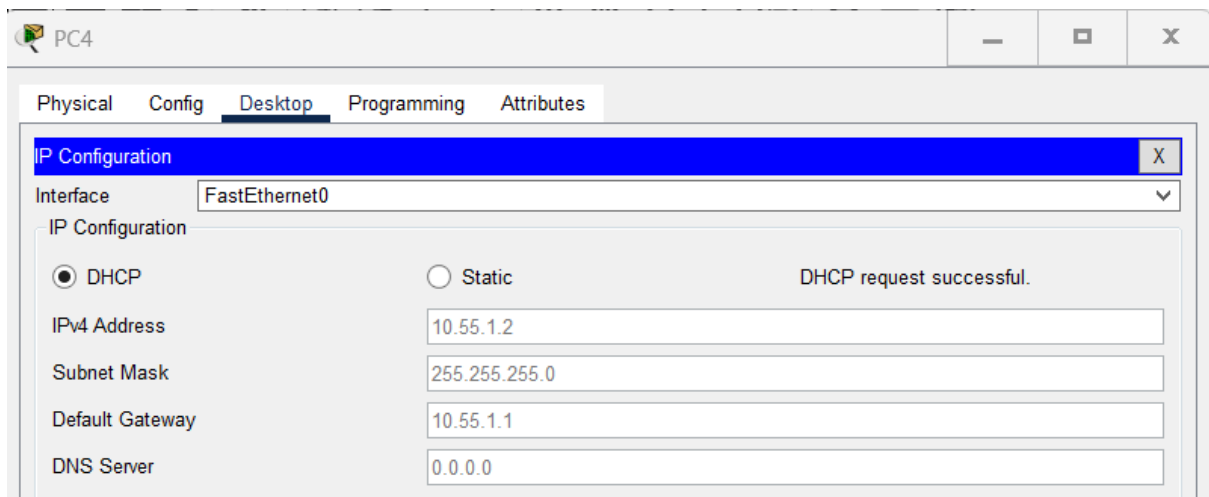


FIGURE 31: FLOOR 1 PC

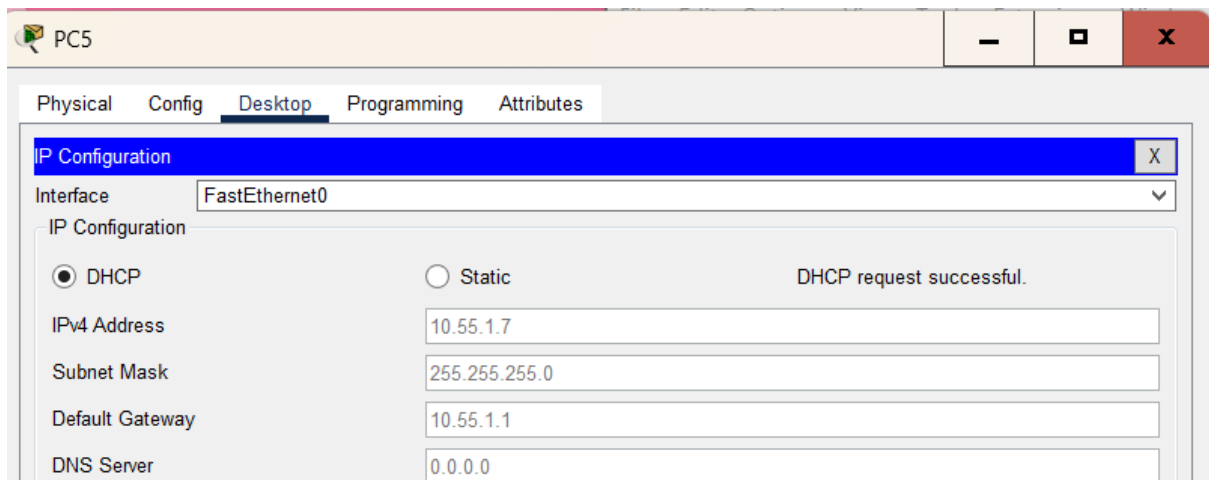


FIGURE 32: FLOOR 2 PC

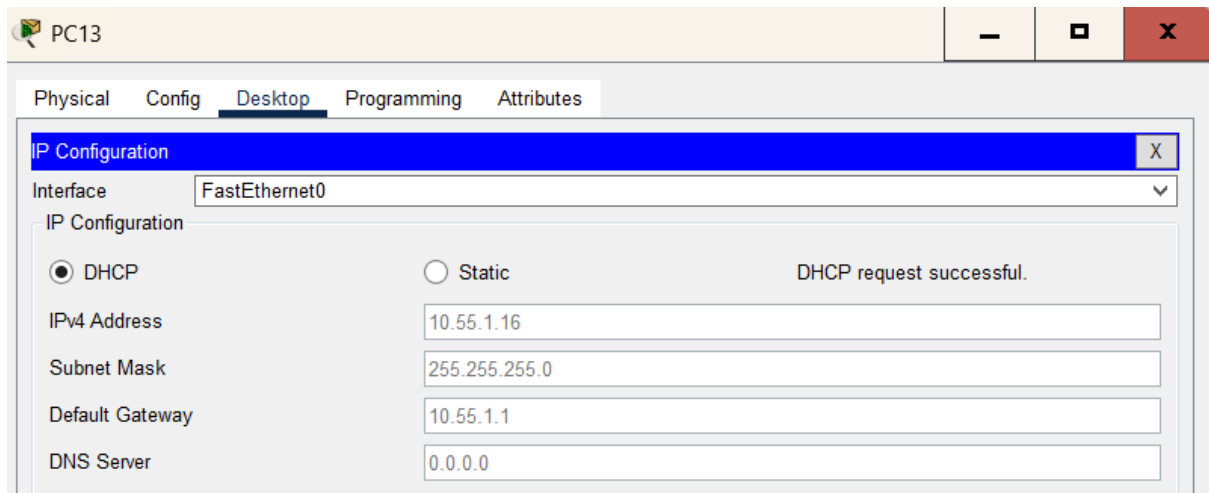


FIGURE 33: FLOOR 3 PC

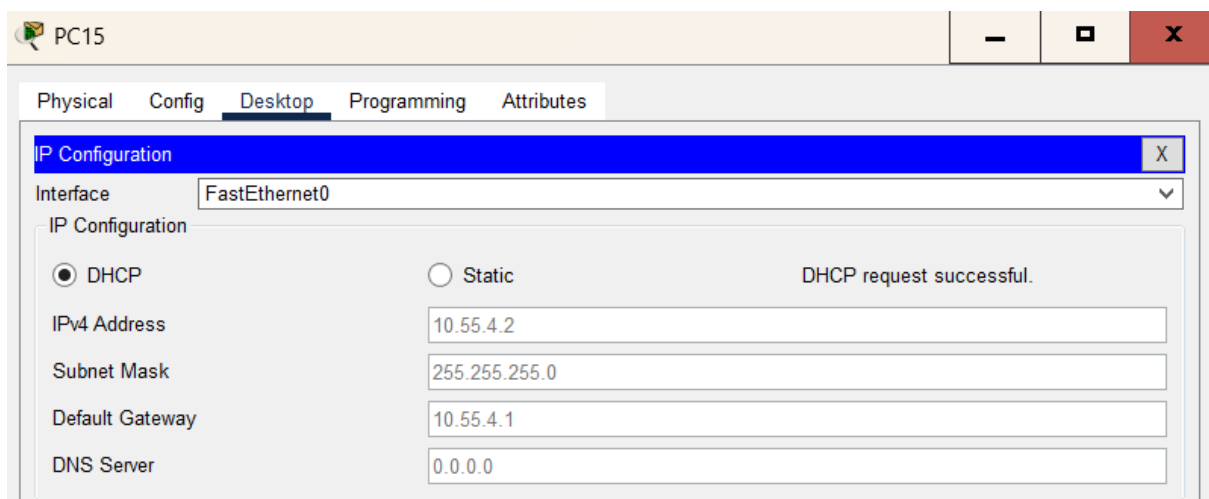


FIGURE 34: FLOOR 4 PC

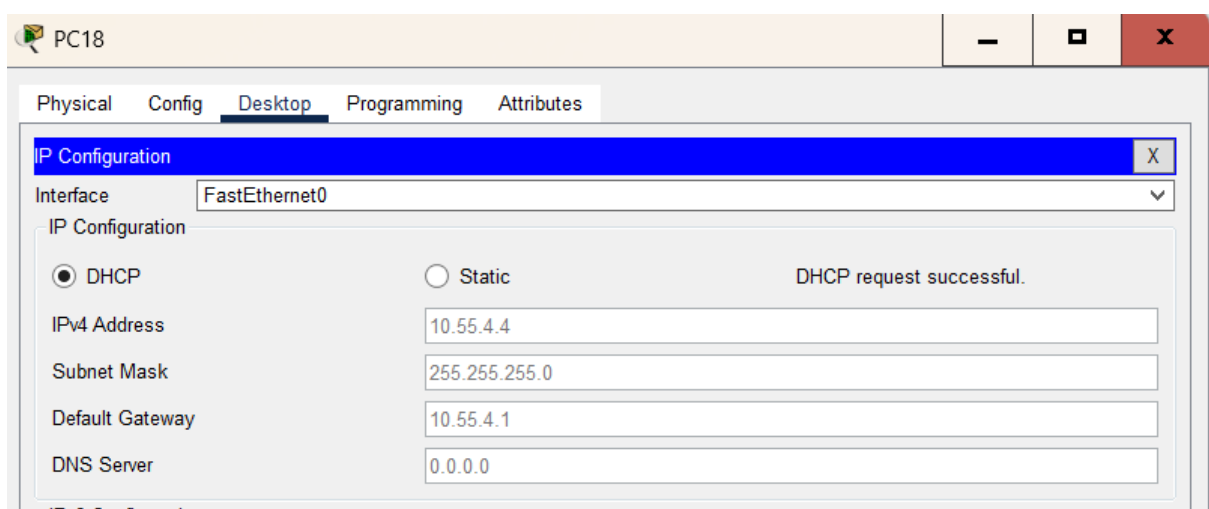


FIGURE 35: FLOOR 4 PC.

```

Switch#show ip dhcp binding
IP address      Client-ID/      Lease expiration    Type
                Hardware address
10.55.1.2       00D0.9759.DA8A   --                  Automatic
10.55.1.3       0009.7CB1.D7CD   --                  Automatic
10.55.1.4       0001.43C4.A912   --                  Automatic
10.55.1.5       0002.1631.CD54   --                  Automatic
10.55.1.6       0060.7054.7343   --                  Automatic
10.55.1.7       000A.F3DA.3448   --                  Automatic
10.55.1.8       0060.3E3E.27E7   --                  Automatic
10.55.1.9       0009.7C9E.9BB7   --                  Automatic
10.55.1.10      000B.BE91.1574   --                  Automatic
10.55.1.11      00D0.5837.E468   --                  Automatic
10.55.1.12      0001.C71C.E4C7   --                  Automatic
10.55.1.13      00D0.9757.643A   --                  Automatic
10.55.1.14      00E0.8F69.03C0   --                  Automatic
10.55.1.15      0001.424A.DCE9   --                  Automatic
10.55.1.16      00E0.A3A3.B906   --                  Automatic
10.55.4.2       0001.6395.D73C   --                  Automatic
10.55.4.3       0060.2F57.7E28   --                  Automatic
10.55.4.4       0090.2B90.DAAC   --                  Automatic
10.55.4.5       0001.C7CC.C467   --                  Automatic
Switch#

```

FIGURE 36: Evidence that DHCP is working across your network

7. Port Security Implementation

The assigned networking feature for this milestone was **Port Security**.

Port Security was implemented on the Access switches on the ports connected directly to client PCs.

It was applied to the PC access ports on Floors 1, 2, 3 and 4. Trunk ports were not configured with Port Security because trunk links are responsible for carrying traffic for multiple VLANs.

The Port Security configuration was designed with the following settings:

- Maximum of one MAC address per protected port
- Sticky MAC address learning
- Restrict violation mode

The use of a maximum of one MAC address means that each protected access port is associated with a single authorised endpoint.

Sticky MAC learning allows the switch to learn the MAC address of the connected device and retain it as a secure MAC address.

Restrict mode was selected so that traffic from an unauthorised MAC address can be restricted without unnecessarily shutting down the entire interface.

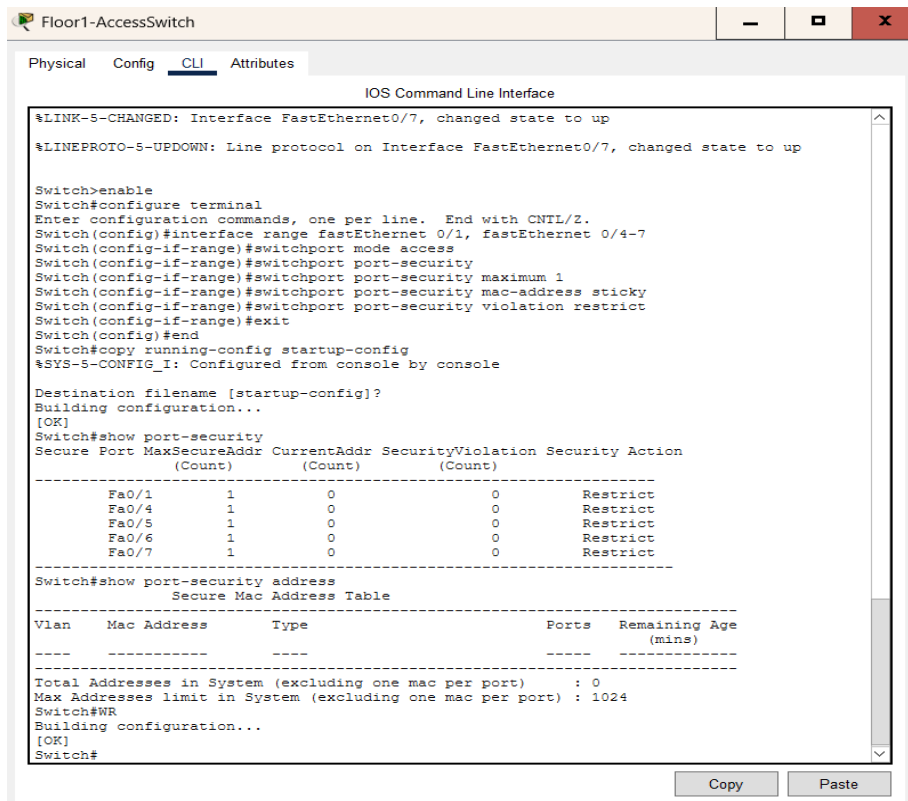


FIGURE37: Port Security Configuration on Floor 1 Access Switch

- **Generate MAC addresses**

All PCs on Floor 1 were used to send traffic, so the **sticky MAC** addresses are learned.

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.55.1.1

Pinging 10.55.1.1 with 32 bytes of data:

Reply from 10.55.1.1: bytes=32 time<1ms TTL=255
Reply from 10.55.1.1: bytes=32 time<1ms TTL=255
Reply from 10.55.1.1: bytes=32 time<1ms TTL=255
Reply from 10.55.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.55.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

```

After pinging all the Floor 1 PCs(Port Security Verification — Floor 1 Staff Access Ports)

Switch#show port-security address

Secure Mac Address Table

Vlan	Mac Address	Type	Ports	Remaining Age (mins)
10	0009.7CB1.D7CD	SecureSticky	Fa0/1	-
10	0002.1631.CD54	SecureSticky	Fa0/4	-
10	0060.7054.7343	SecureSticky	Fa0/5	-
10	0001.43C4.A912	SecureSticky	Fa0/6	-
10	00D0.9759.DA8A	SecureSticky	Fa0/7	-

Total Addresses in System (excluding one mac per port) : 0

Max Addresses limit in System (excluding one mac per port) : 1024

Switch#

Floor2-AccessSwitch

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fastEthernet 0/1, fastEthernet 0/4-7
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport port-security
Switch(config-if-range)#switchport port-security maximum 1
Switch(config-if-range)#switchport port-security mac-address sticky
Switch(config-if-range)#switchport port-security violation restrict
Switch(config-if-range)#exit
Switch(config)#end
Switch#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
      (Count)      (Count)      (Count)
-----
Fa0/4          1          0          0          Restrict
Fa0/5          1          0          0          Restrict
Fa0/6          1          0          0          Restrict
Fa0/7          1          0          0          Restrict

Switch#show port-security address
Secure Mac Address Table
Vlan  Mac Address      Type      Ports  Remaining Age
-----
10    0009.7CB1.D7CD    SecureSticky  Fa0/1  -
10    0002.1631.CD54    SecureSticky  Fa0/4  -
10    0060.7054.7343    SecureSticky  Fa0/5  -
10    0001.43C4.A912    SecureSticky  Fa0/6  -
10    00D0.9759.DA8A    SecureSticky  Fa0/7  -

Total Addresses in System (excluding one mac per port) : 0
Max Addresses limit in System (excluding one mac per port) : 1024
Switch#
```

Port Security Configuration on Floor 2 Access Switch

ALL PCs on Floor 2 were used to send traffic, so the **sticky MAC** addresses are learned

PC7

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.55.1.1

Pinging 10.55.1.1 with 32 bytes of data:

Reply from 10.55.1.1: bytes=32 time=32ms TTL=255
Reply from 10.55.1.1: bytes=32 time<1ms TTL=255
Reply from 10.55.1.1: bytes=32 time<1ms TTL=255
Reply from 10.55.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.55.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 32ms, Average = 8ms

C:\>
```

After pinging all the Floor 1 PCs(Port Security Verification — Floor 1 Staff Access Ports)

```
Switch#show port-security address
Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
(mins)
-----
10      000A.F3DA.3448   SecureSticky        Fa0/1    -
10      0060.3E3E.27E7   SecureSticky        Fa0/4    -
10      0009.7C9E.9BB7   SecureSticky        Fa0/5    -
10      000B.BE91.1574   SecureSticky        Fa0/6    -
10      00D0.5837.E468   SecureSticky        Fa0/7    -
-----
Total Addresses in System (excluding one mac per port)    : 0
Max Addresses limit in System (excluding one mac per port) : 1024
Switch#wr
Building configuration...
[OK]
Switch#
```

FLOOR 3

Floor3-AccessSwitch

Physical

Config

CLI

Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/6, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/7, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface range fastEthernet 0/1, fastEthernet 0/4-7
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport port-security
Switch(config-if-range)#switchport port-security maximum 1
Switch(config-if-range)#switchport port-security mac-address sticky
Switch(config-if-range)#switchport port-security violation restrict
Switch(config-if-range)#exit
Switch(config)#end
Switch#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

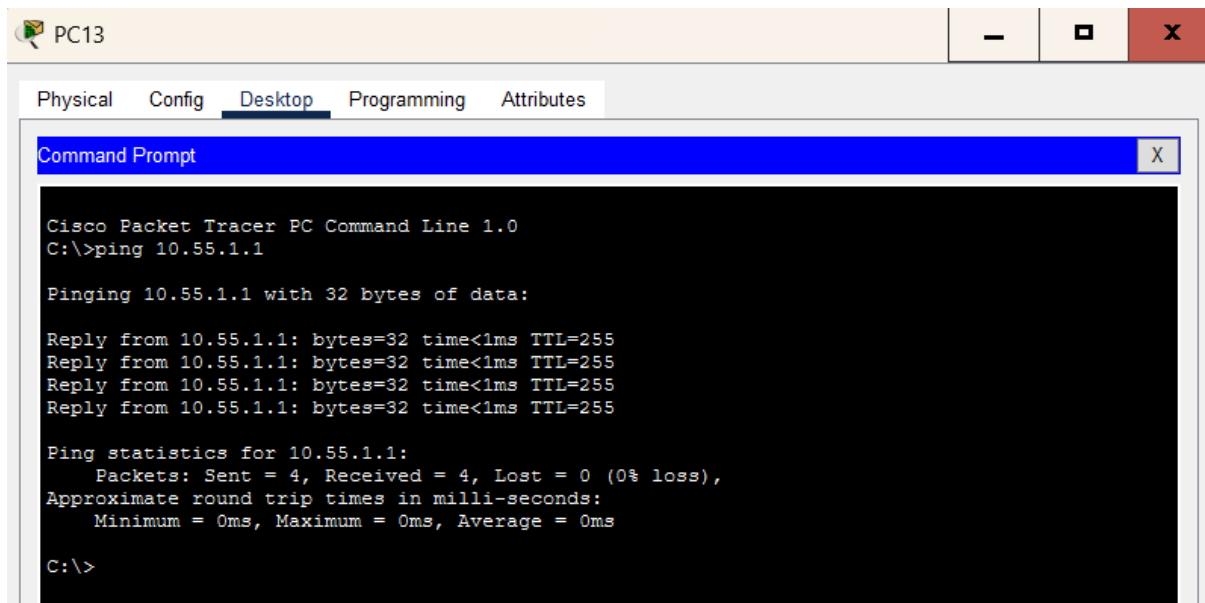
Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
      (Count)      (Count)      (Count)
-----
Fa0/1          1          0          0          Restrict
Fa0/4          1          0          0          Restrict
Fa0/5          1          0          0          Restrict
Fa0/6          1          0          0          Restrict
Fa0/7          1          0          0          Restrict
-----
Switch#show port-security address
Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
(mins)
-----
-----
Total Addresses in System (excluding one mac per port)    : 0
Max Addresses limit in System (excluding one mac per port) : 1024
Switch#
```

Copy

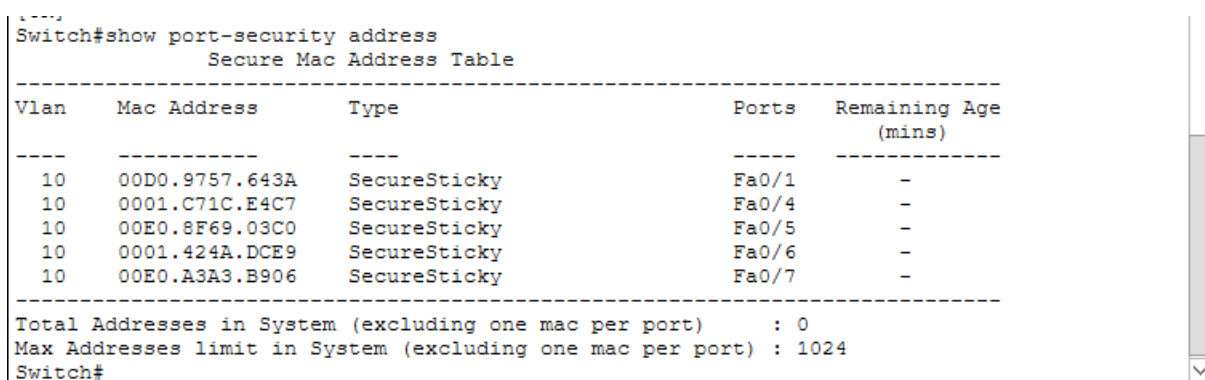
Paste

☐ Top

ALL PCs on Floor 3 were used to send traffic, so the **sticky MAC** addresses are learned



Floor 3 Port Security Verification



FLOOR 4

Floor4-AccessSwitch

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fastEthernet 0/1, fastEthernet 0/4-6
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport port-security
Switch(config-if-range)#switchport port-security maximum 1
Switch(config-if-range)#switchport port-security mac-address sticky
Switch(config-if-range)#switchport port-security violation restrict
Switch(config-if-range)#exit
Switch(config)#end
Switch#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show port-security
```

Secure Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
Fa0/1	1	0	0	Restrict
Fa0/4	1	0	0	Restrict
Fa0/5	1	0	0	Restrict
Fa0/6	1	0	0	Restrict

```
Switch#show port-security address
Secure Mac Address Table
```

Vlan	Mac Address	Type	Ports	Remaining Age (mins)
----	-----	----	----	-----

```
Total Addresses in System (excluding one mac per port) : 0
Max Addresses limit in System (excluding one mac per port) : 1024
```

ALL PCs on Floor 4 were used to send traffic, so the **sticky MAC** addresses are learned

PC17

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.55.4.1

Pinging 10.55.4.1 with 32 bytes of data:

Reply from 10.55.4.1: bytes=32 time=21ms TTL=255
Reply from 10.55.4.1: bytes=32 time<1ms TTL=255
Reply from 10.55.4.1: bytes=32 time<1ms TTL=255
Reply from 10.55.4.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.55.4.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 21ms, Average = 5ms

C:\>
```

Floor 4 Port Security Verification-Management Access Ports

```
Switch#show port-security address
      Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
      (mins)
-----
  40     0001.6395.D73C   SecureSticky        Fa0/1     -
  40     0001.C7CC.C467   SecureSticky        Fa0/4     -
  40     0060.2F57.7E28   SecureSticky        Fa0/5     -
  40     0090.2B90.DAAC   SecureSticky        Fa0/6     -
-----
Total Addresses in System (excluding one mac per port)    : 0
Max Addresses limit in System (excluding one mac per port) : 1024
Switch#
```

Port Security Configuration and Status — Floor 4 Fa0/1

```
Switch#show port-security address
      Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
      (mins)
-----
  40     0001.6395.D73C   SecureSticky        Fa0/1     -
  40     0001.C7CC.C467   SecureSticky        Fa0/4     -
  40     0060.2F57.7E28   SecureSticky        Fa0/5     -
  40     0090.2B90.DAAC   SecureSticky        Fa0/6     -
-----
Total Addresses in System (excluding one mac per port)    : 0
Max Addresses limit in System (excluding one mac per port) : 1024
Switch#wr
Building configuration...
[OK]
Switch#show port-security interface fa0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Restrict
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 1
Total MAC Addresses    : 1
Configured MAC Addresses : 0
Sticky MAC Addresses   : 1
Last Source Address:Vlan : 0001.6395.D73C:40
Security Violation Count : 0
```

After configuring Port Security, the client devices were connected, and traffic was generated to allow the switch to learn their MAC addresses.

The Port Security configuration was then verified on the Access switches.

The verification confirmed that the protected ports were operating as secure ports and that the expected sticky MAC addresses had been learned.

- **Why Port Security is Appropriate**

Port Security is appropriate for this network because the Access switches provide direct physical connections to end-user devices. Without port security, an unauthorised device could potentially be connected to an available network port and attempt to gain access to the network. Restricting each client port to one MAC address reduces this risk by limiting the number of devices that can use the port. Sticky MAC addresses also make the configuration practical for the implemented network because the switch can learn the MAC address of the intended client device rather than requiring every MAC address to be manually entered. The feature therefore provides an additional layer of security at the Access layer of the network.

- **Why Port Security is Appropriate**

Port Security is appropriate for this network because the Access switches provide direct physical connections to end-user devices. Without port security, an unauthorised device could potentially be connected to an available network port and attempt to gain access to the network. Restricting each client port to one MAC address reduces this risk by limiting the number of devices that can use the port. Sticky MAC addresses also make the configuration practical for the implemented network because the switch can learn the MAC address of the intended client device rather than requiring every MAC address to be manually entered. The feature therefore provides an additional layer of security at the Access layer of the network.

8. Staff VLAN Connectivity Verification

To verify network connectivity within the Staff VLAN, ICMP echo requests (ping tests) were performed from a host device to both the default gateway and another device located in the same subnet. The first test targeted the Staff VLAN gateway address (**10.55.1.1**), while the second test targeted a Staff VLAN workstation (**10.55.1.17**).

The results indicate that all transmitted packets were successfully received with **0% packet loss**, confirming that the VLAN configuration, access port assignments, and Layer 3 gateway functionality are operating correctly. Communication with the default gateway verifies that end devices can reach the VLAN's routing interface, while successful communication with the workstation confirms proper host-to-host connectivity within the Staff VLAN.

The recorded round-trip times (RTT) demonstrate normal network performance, with responses received within a few milliseconds. These results confirm that devices within the Staff VLAN can communicate effectively and that the network infrastructure has been configured correctly.

Successful ping tests from a Staff VLAN host to the default gateway (10.55.1.1) and another Staff VLAN device (10.55.1.17). All packets were transmitted and received successfully with 0% packet loss, confirming proper VLAN operation, gateway accessibility, and intra-VLAN communication.

```

C:\>
ping 10.55.1.1

Pinging 10.55.1.1 with 32 bytes of data:

Reply from 10.55.1.1: bytes=32 time=43ms TTL=255
Reply from 10.55.1.1: bytes=32 time=82ms TTL=255
Reply from 10.55.1.1: bytes=32 time=28ms TTL=255
Reply from 10.55.1.1: bytes=32 time=33ms TTL=255

Ping statistics for 10.55.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 28ms, Maximum = 82ms, Average = 46ms

C:\>ping 10.55.1.17

Pinging 10.55.1.17 with 32 bytes of data:

Reply from 10.55.1.17: bytes=32 time<1ms TTL=128
Reply from 10.55.1.17: bytes=32 time=21ms TTL=128
Reply from 10.55.1.17: bytes=32 time=32ms TTL=128
Reply from 10.55.1.17: bytes=32 time=30ms TTL=128

Ping statistics for 10.55.1.17:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 32ms, Average = 20ms

```

9. Router Configuration

Router-to-Core Interface Verification — ISR4331 G0/0/0

```

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0/0
Router(config-if)#ip address 10.55.254.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0/0	10.55.254.2	YES	manual	up	up
GigabitEthernet0/0/1	unassigned	YES	unset	administratively down	down
GigabitEthernet0/0/2	unassigned	YES	unset	administratively down	down

```

Router#

```

Core interface verification.

Core-3650

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface gigabitEthernet 1/0/1
Switch(config-if)#no switchport
Switch(config-if)#ip address 10.55.254.1 255.255.255.252
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet1/0/1, changed state to up

Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet1/0/1     10.55.254.1     YES manual up          up
GigabitEthernet1/0/2     unassigned      YES unset up          up
GigabitEthernet1/0/3     unassigned      YES unset up          up
GigabitEthernet1/0/4     unassigned      YES unset up          up
GigabitEthernet1/0/5     unassigned      YES unset up          up
GigabitEthernet1/0/6     unassigned      YES unset down        down
GigabitEthernet1/0/7     unassigned      YES unset down        down
GigabitEthernet1/0/8     unassigned      YES unset down        down
GigabitEthernet1/0/9     unassigned      YES unset down        down
GigabitEthernet1/0/10    unassigned      YES unset down        down
GigabitEthernet1/0/11    unassigned      YES unset down        down
GigabitEthernet1/0/12    unassigned      YES unset down        down
GigabitEthernet1/0/13    unassigned      YES unset down        down
GigabitEthernet1/0/14    unassigned      YES unset down        down
GigabitEthernet1/0/15    unassigned      YES unset down        down
GigabitEthernet1/0/16    unassigned      YES unset down        down
GigabitEthernet1/0/17    unassigned      YES unset down        down
GigabitEthernet1/0/18    unassigned      YES unset down        down
GigabitEthernet1/0/19    unassigned      YES unset down        down
GigabitEthernet1/0/20    unassigned      YES unset down        down
GigabitEthernet1/0/21    unassigned      YES unset down        down
GigabitEthernet1/0/22    unassigned      YES unset down        down
GigabitEthernet1/0/23    unassigned      YES unset down        down
GigabitEthernet1/0/24    unassigned      YES unset down        down
GigabitEthernet1/1/1     unassigned      YES unset down        down
--More--
```

Router-Core connectivity is verified.

```
Router#ping 10.55.254.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.254.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

-----
Switch#ping 10.55.254.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.254.2, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/2 ms

Switch#wr
Building configuration...
Compressed configuration from 7383 bytes to 3601 bytes[OK]
[OK]
Switch#
```

To enable communication between the external router and the internal enterprise network, static routes were configured on the router for all VLAN subnets. Each route directs traffic destined for the Staff (10.55.1.0/24), Guest (10.55.2.0/24), Server (10.55.3.0/24), and Management (10.55.4.0/24) networks through the next-hop address **10.55.254.1**, which represents the Layer 3 Core Switch interface.

After configuration, the show IP route command was used to verify the routing table. The output confirms that all four VLAN networks have been successfully installed as static routes, indicated by the **S** route code. The routing table also shows that the point-to-point network **10.55.254.0/30** is directly connected to the router through interface GigabitEthernet0/0/0.

This configuration ensures that the router can forward traffic to all internal VLANs via the Core Switch, enabling connectivity between external networks and the internal enterprise network infrastructure.

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#ip route 10.55.1.0 255.255.255.0 10.55.254.1
Router(config)#ip route 10.55.2.0 255.255.255.0 10.55.254.1
Router(config)#ip route 10.55.3.0 255.255.255.0 10.55.254.1
Router(config)#ip route 10.55.4.0 255.255.255.0 10.55.254.1
Router(config)#
Router(config)#end
Router#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

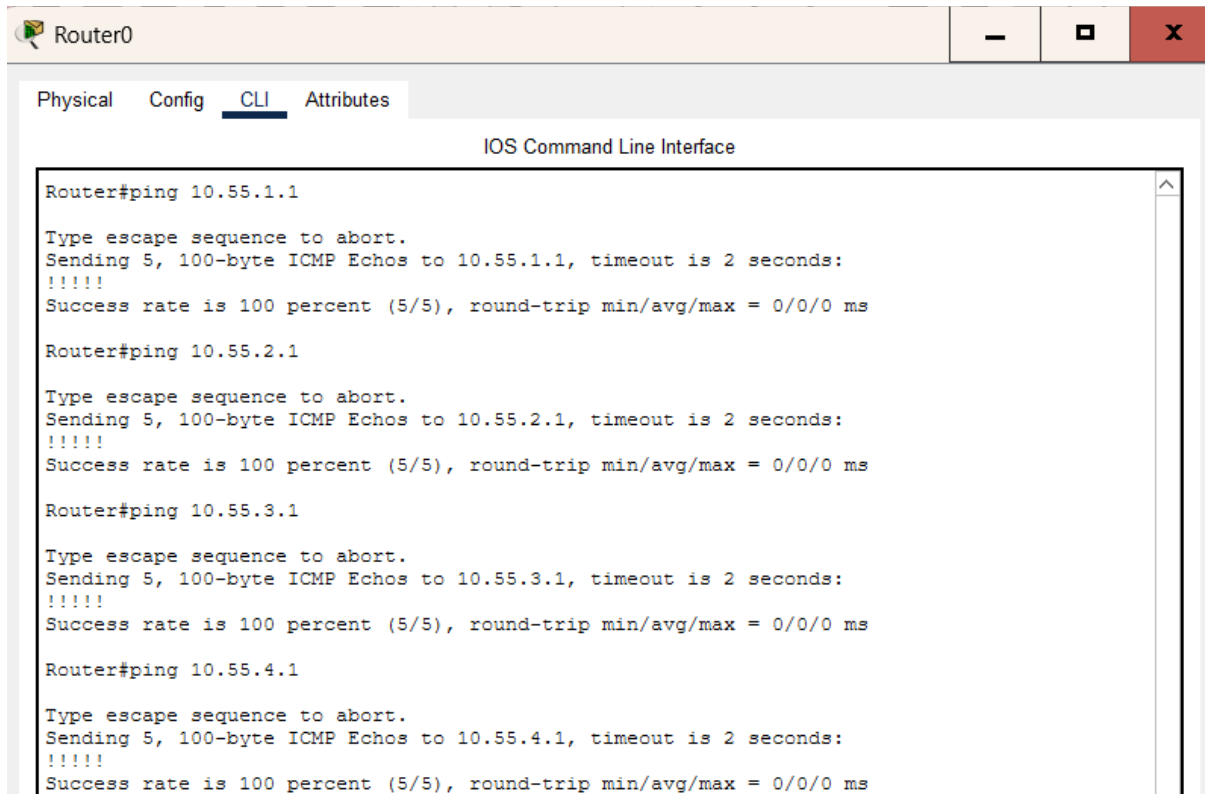
Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 3 masks
S       10.55.1.0/24 [1/0] via 10.55.254.1
S       10.55.2.0/24 [1/0] via 10.55.254.1
S       10.55.3.0/24 [1/0] via 10.55.254.1
S       10.55.4.0/24 [1/0] via 10.55.254.1
C       10.55.254.0/30 is directly connected, GigabitEthernet0/0/0
L       10.55.254.2/32 is directly connected, GigabitEthernet0/0/0
```

Static Route Configuration and Verification on the Router

- **Router Connectivity Test to All Internal VLAN Gateways**

Successful ping tests from the router to the Staff (10.55.1.1), Guest (10.55.2.1), Server (10.55.3.1), and Management (10.55.4.1) VLAN gateway interfaces. The 100% success rate confirms correct static route configuration and reachability of all internal VLAN networks through the Core Layer 3 switch.



```
Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router#ping 10.55.1.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.1.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 10.55.2.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.2.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 10.55.3.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.3.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms

Router#ping 10.55.4.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.55.4.1, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/0 ms
```

Guest VLAN Implementation

A dedicated Guest VLAN was implemented using **VLAN 20**.

The Guest network uses **10.55.2.0/24**, with **10.55.2.1** configured as the default gateway.

Guest PCs were added to Floors 1, 2 and 3 and connected to dedicated access ports configured for VLAN 20.

This provided a clear separation between Guest users and the Staff, Server and Management networks.

Guest Isolation

Guest users were required to be prevented from accessing internal network resources.

An extended access control list named **GUEST_ISOLATION** was implemented on the Core switch.

The ACL prevents traffic originating from the Guest network from reaching:

- Staff network — 10.55.1.0/24
- Server network — 10.55.3.0/24
- Management network — 10.55.4.0/24

Guest traffic that is not destined for these internal networks is permitted.

The ACL was applied to the Guest VLAN interface so that Guest traffic is filtered when it enters the Core switch for routing.

GUEST ISOLATION CONFIGURATION and ACCESS LIST

```
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#
Switch(config)#ip access-list extended GUEST_ISOLATION
Switch(config-ext-nacl)#deny ip 10.55.2.0 0.0.0.255 10.55.1.0 0.0.0.255
Switch(config-ext-nacl)#deny ip 10.55.2.0 0.0.0.255 10.55.3.0 0.0.0.255
Switch(config-ext-nacl)#deny ip 10.55.2.0 0.0.0.255 10.55.4.0 0.0.0.255
Switch(config-ext-nacl)#permit ip 10.55.2.0 0.0.0.255 any
Switch(config-ext-nacl)#exit
Switch(config)#
Switch(config)#interface vlan 20
Switch(config-if)#ip access-group GUEST_ISOLATION in
Switch(config-if)#exit
Switch(config)#
Switch(config)#end
Switch#copy running-config startup-config
%SYS-5-CONFIG_I: Configured from console by console

Destination filename [startup-config]?
Building configuration...
[OK]
Switch#show access-lists
Extended IP access list GUEST_ISOLATION
 10 deny ip 10.55.2.0 0.0.0.255 10.55.1.0 0.0.0.255
 20 deny ip 10.55.2.0 0.0.0.255 10.55.3.0 0.0.0.255
 30 deny ip 10.55.2.0 0.0.0.255 10.55.4.0 0.0.0.255
 40 permit ip 10.55.2.0 0.0.0.255 any
Switch#
```

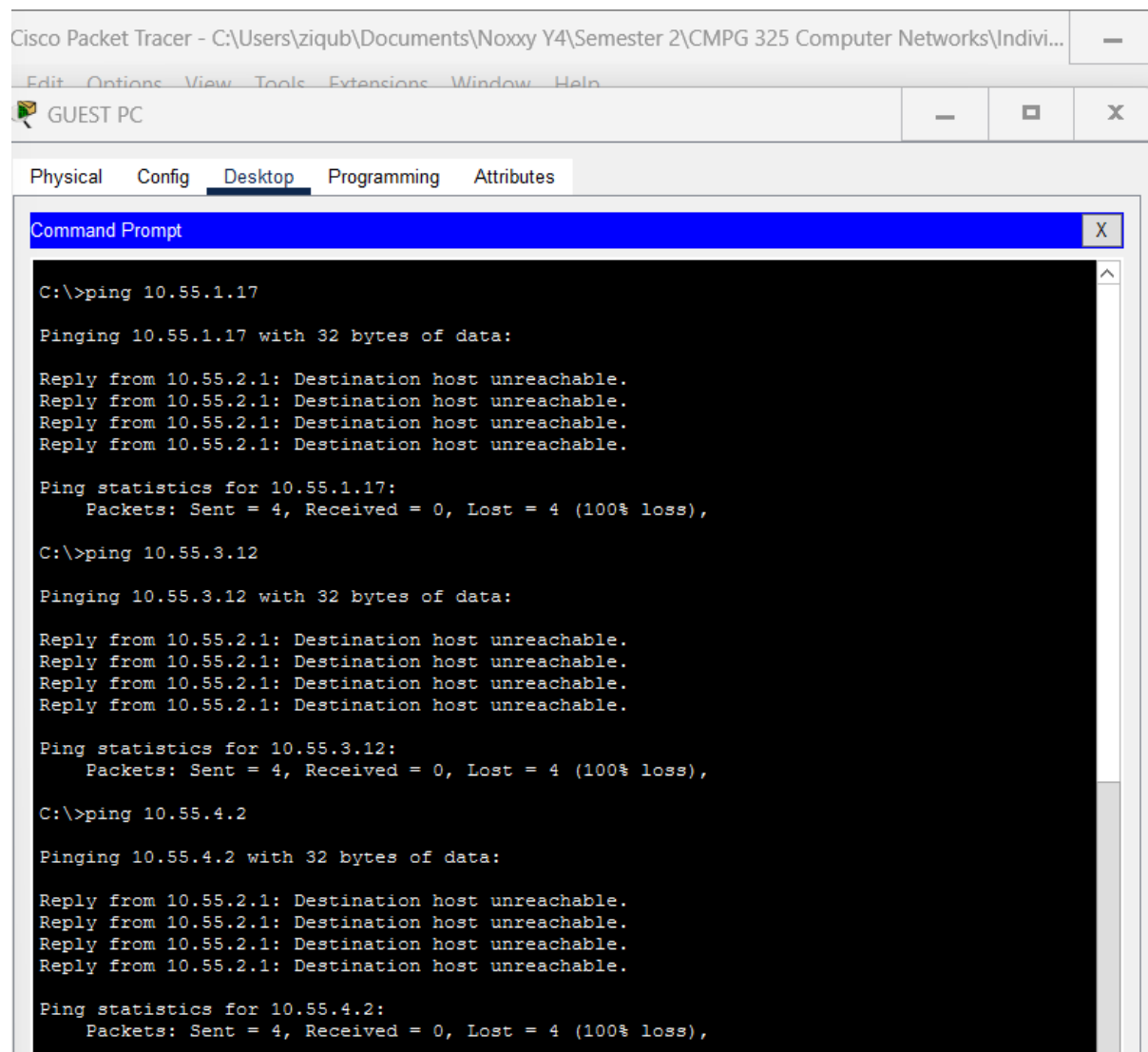
Guest VLAN Isolation Access Control List (ACL)

Verification of the GUEST_ISOLATION extended ACL configured on the Layer 3 switch. The ACL denies Guest VLAN (10.55.2.0/24) access to the Staff, Server, and Management VLANs while permitting access to all other networks. Match counters confirm that unauthorized traffic attempts were successfully filtered

```
Switch# show access-lists GUEST_ISOLATION
Extended IP access list GUEST_ISOLATION
  deny ip 10.55.2.0 0.0.0.255 10.55.1.0 0.0.0.255 (4 match(es))
  deny ip 10.55.2.0 0.0.0.255 10.55.3.0 0.0.0.255 (4 match(es))
  deny ip 10.55.2.0 0.0.0.255 10.55.4.0 0.0.0.255 (4 match(es))
  permit ip 10.55.2.0 0.0.0.255 any (4 match(es))
```

Guest VLAN Access Restriction Verification

Ping tests performed from a Guest VLAN host (10.55.2.0/24) to devices in the Staff, Server, and Management VLANs. All requests returned "Destination host unreachable" with 100% packet loss, confirming that the GUEST_ISOLATION ACL successfully blocks unauthorized inter-VLAN communication.



10. Conclusion

The enterprise network was successfully implemented in Cisco Packet Tracer using a hierarchical design consisting of a Core switch, Distribution switches, Access switches and an edge router. VLAN segmentation, trunking, inter-VLAN routing, DHCP, static routing, Guest VLAN isolation and Port Security were configured and tested. Verification results confirmed successful connectivity between authorised devices, proper VLAN segregation, automatic IP address allocation, secure access port operation and effective restriction of Guest network access. The implemented network satisfies the project requirements and demonstrates a scalable, secure and functional enterprise network solution